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(54) **WORKING MACHINE, INFORMATION
PROCESSING DEVICE, AND PROGRAM**

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(57) **ABSTRACT**

An information processing device includes an operation planning part configured to determine a future operation of a working machine from among a plurality of operations according to a performance status of an operation of the working machine. A non-transitory computer-readable recording medium has a program embodied therein for causing an information processing device to determine a future operation of a working machine from among a plurality of operations according to a performance status of an operation of the working machine.

SYS

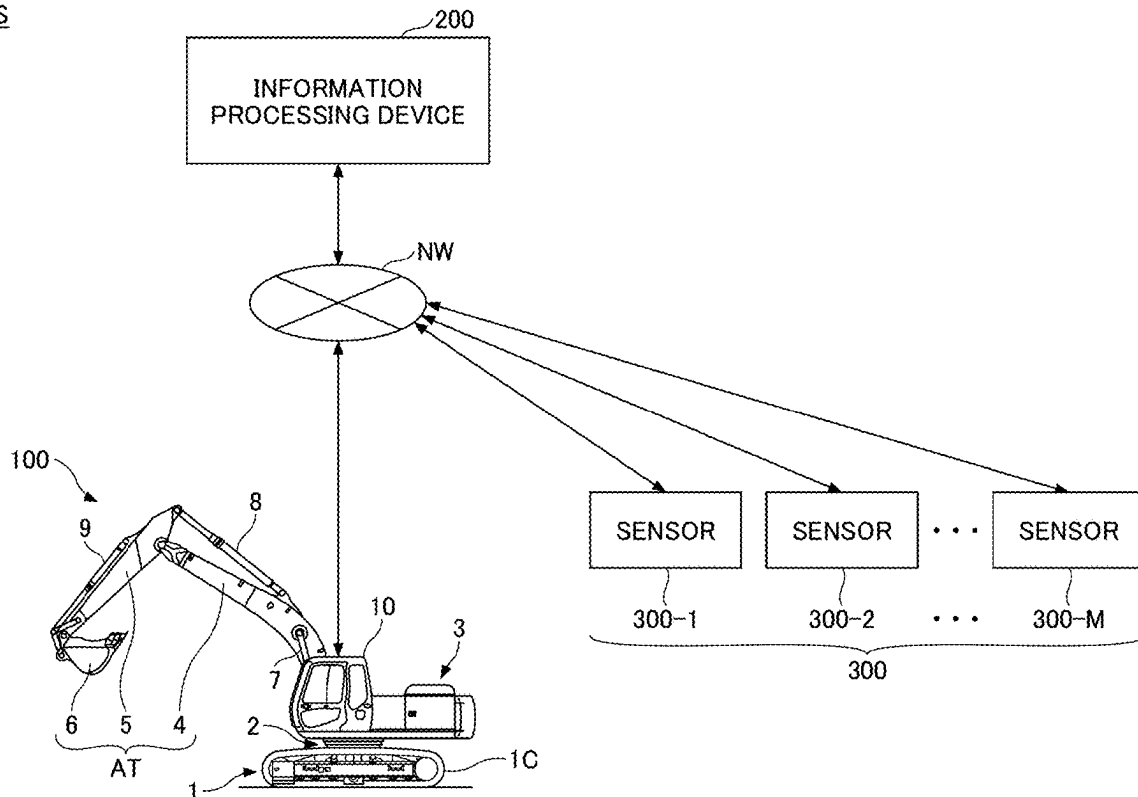


FIG.1

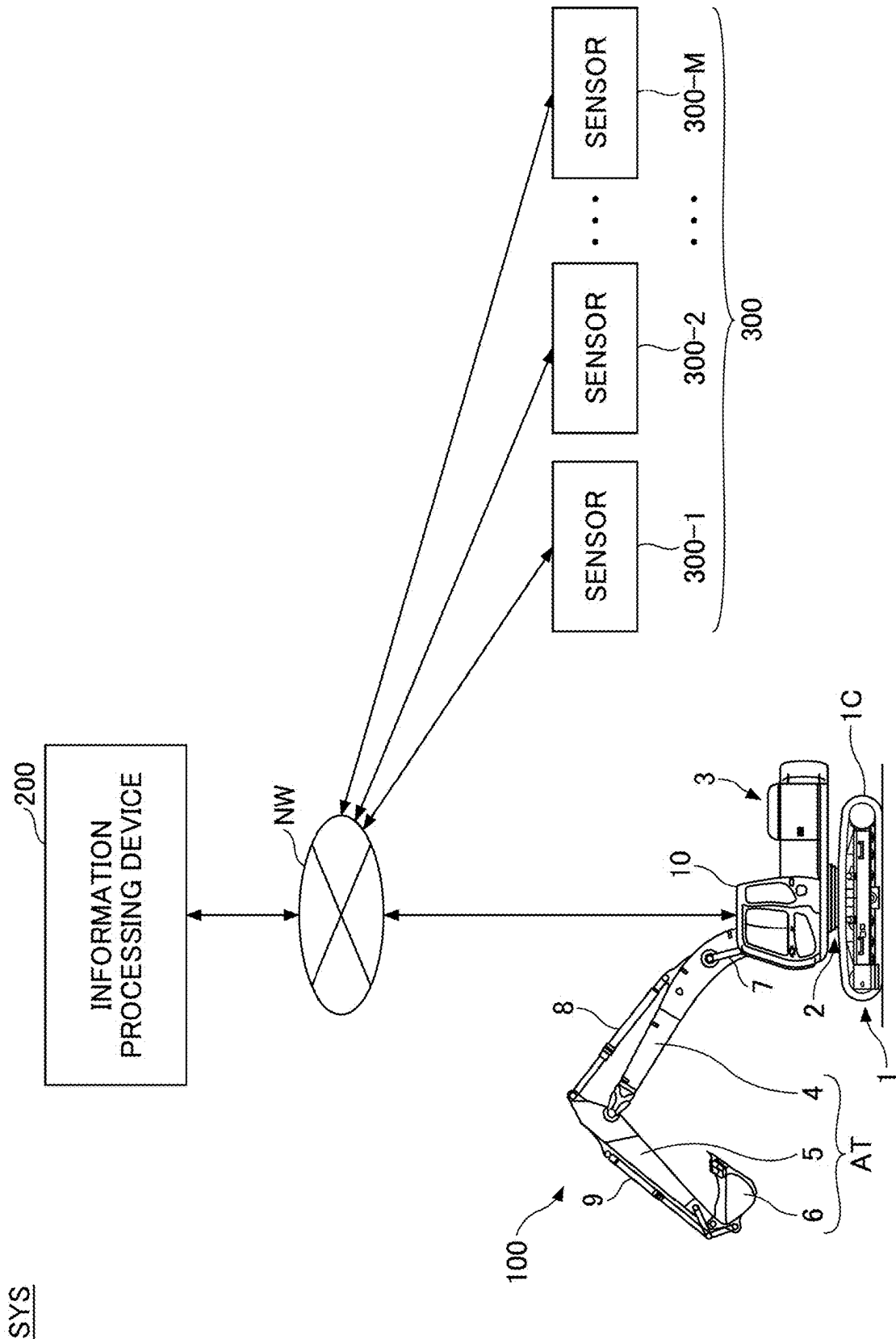


FIG.3

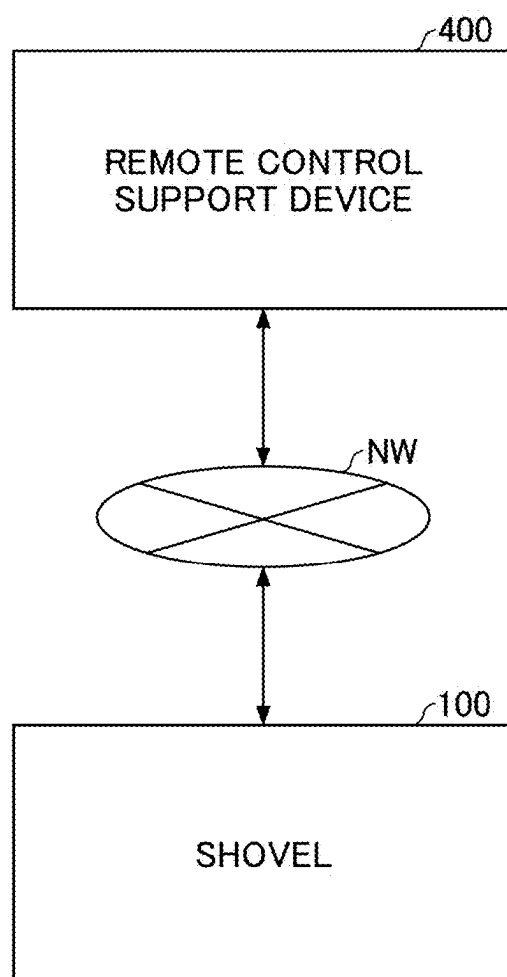


FIG.4

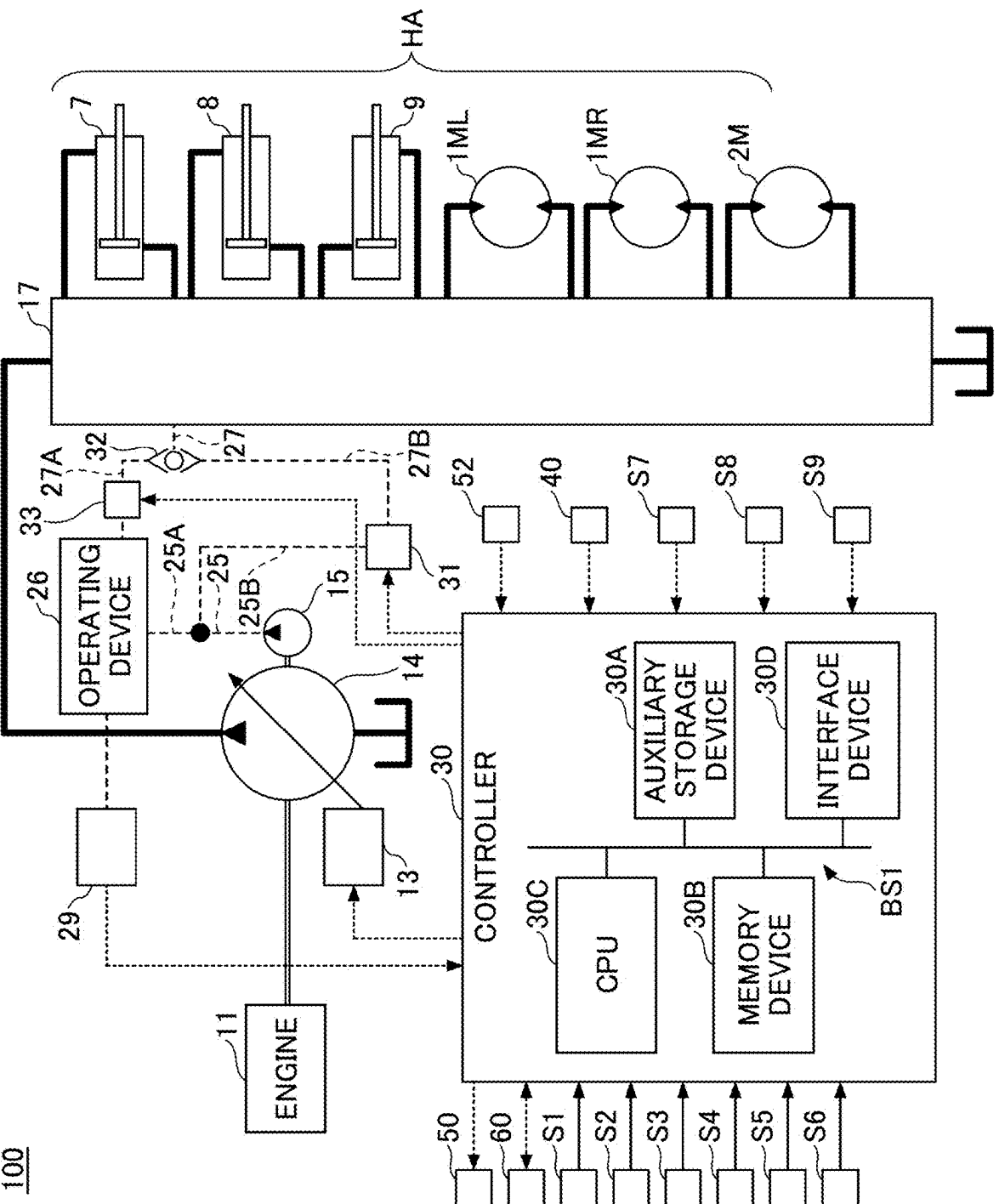


FIG.5

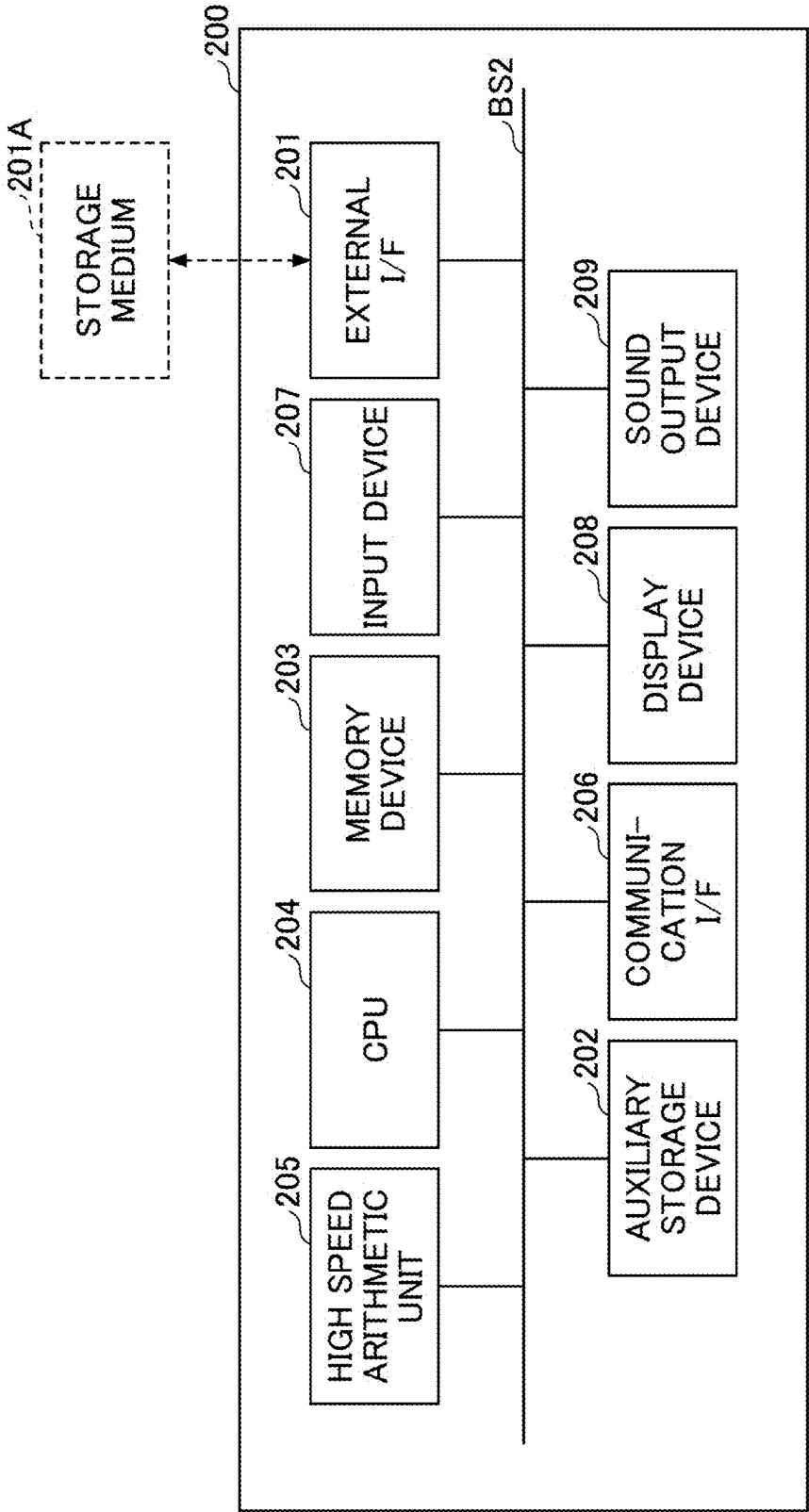


FIG. 6

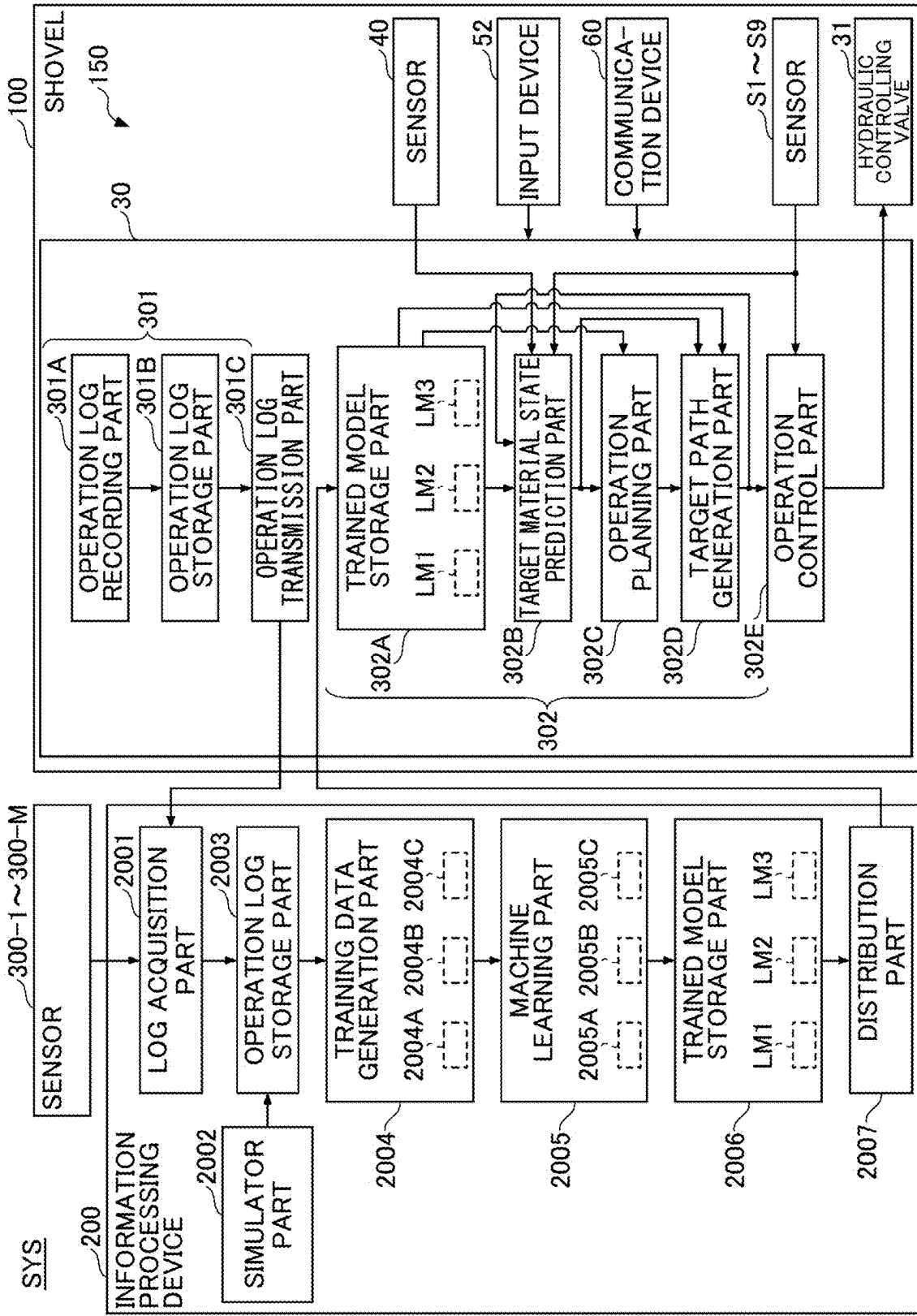
SYS

FIG.7

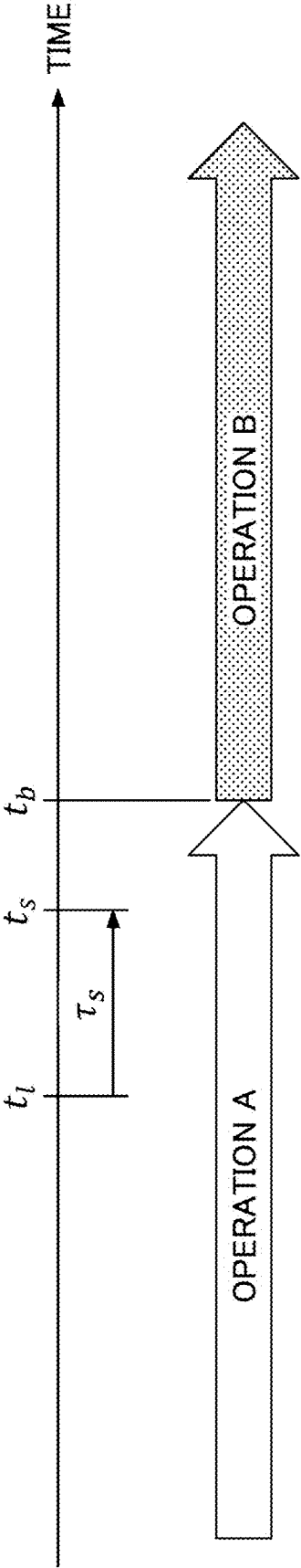


FIG.8

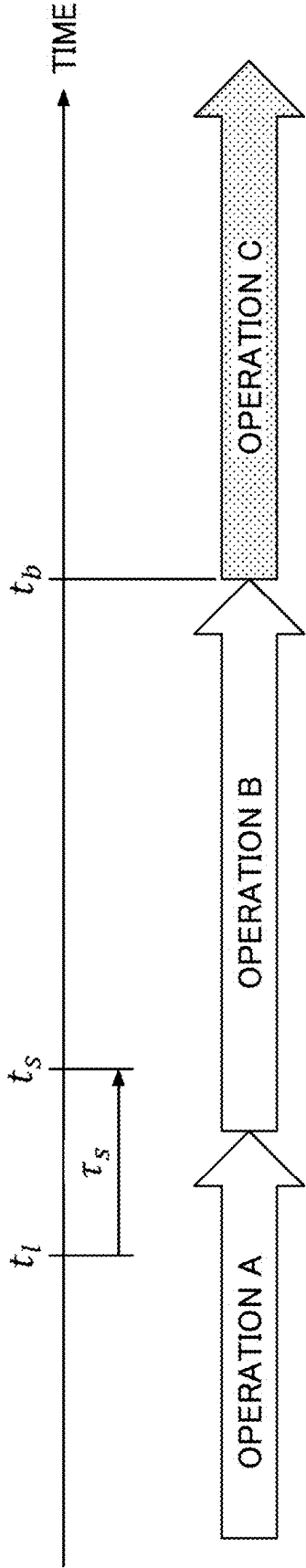


FIG.9

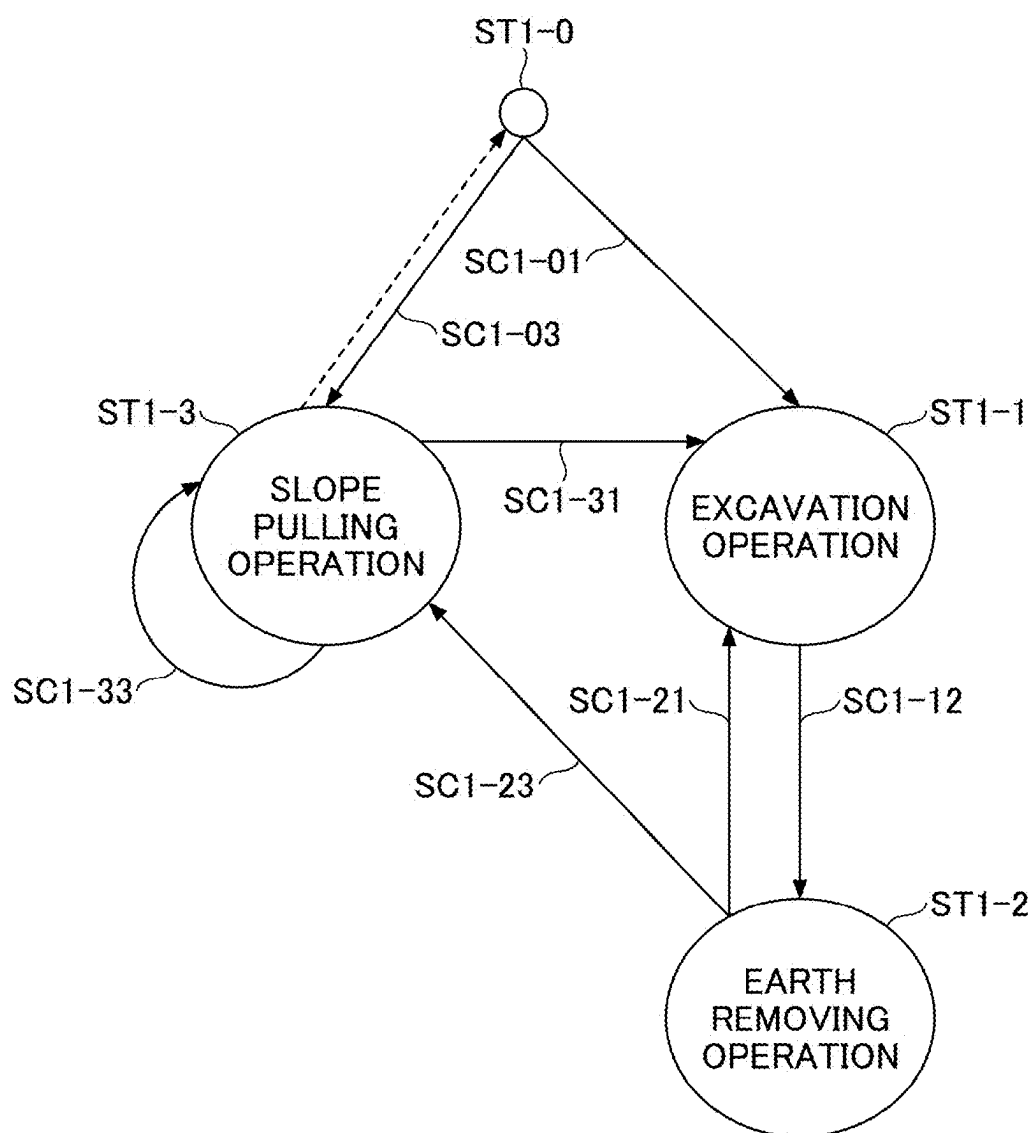
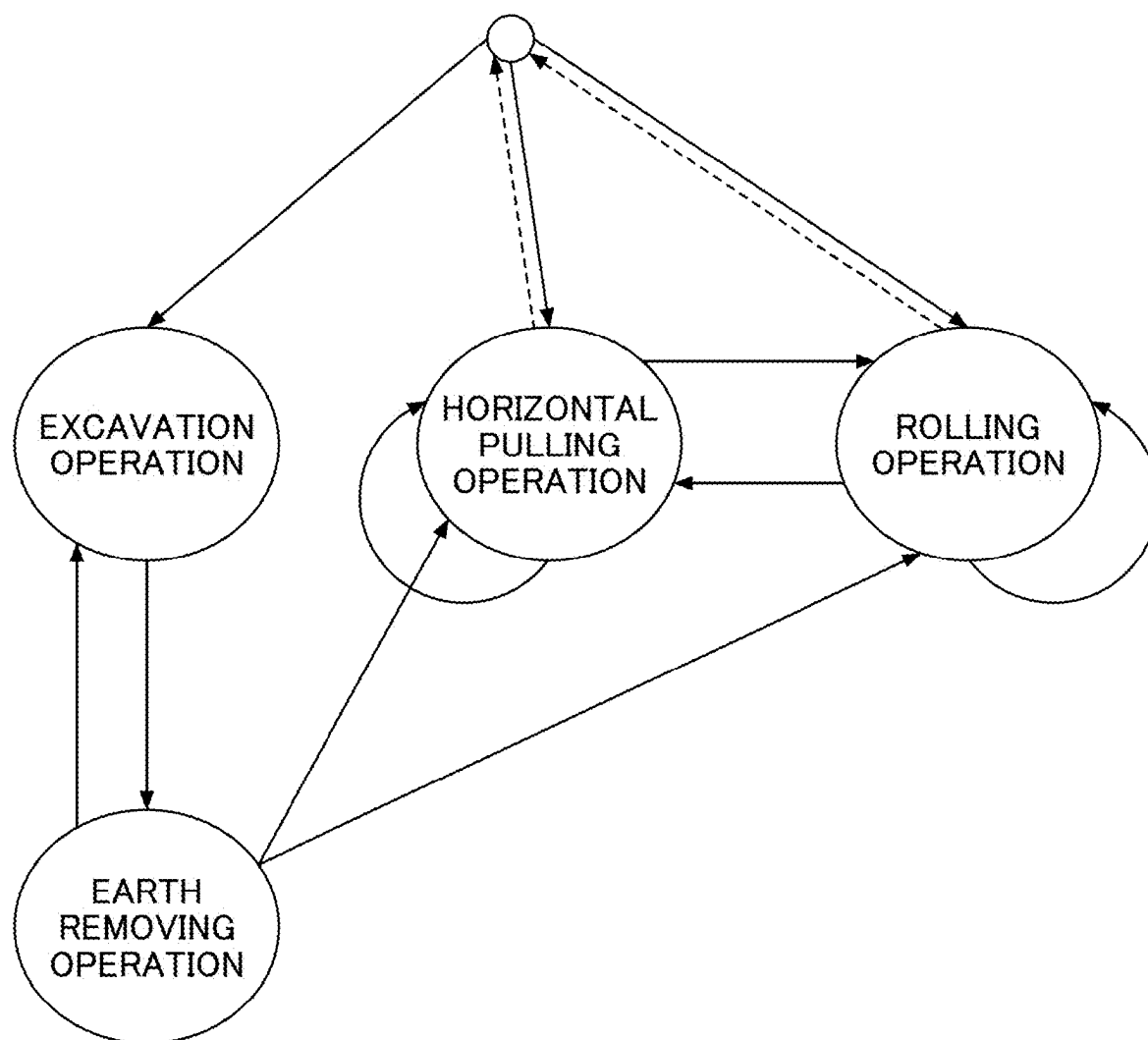


FIG.10



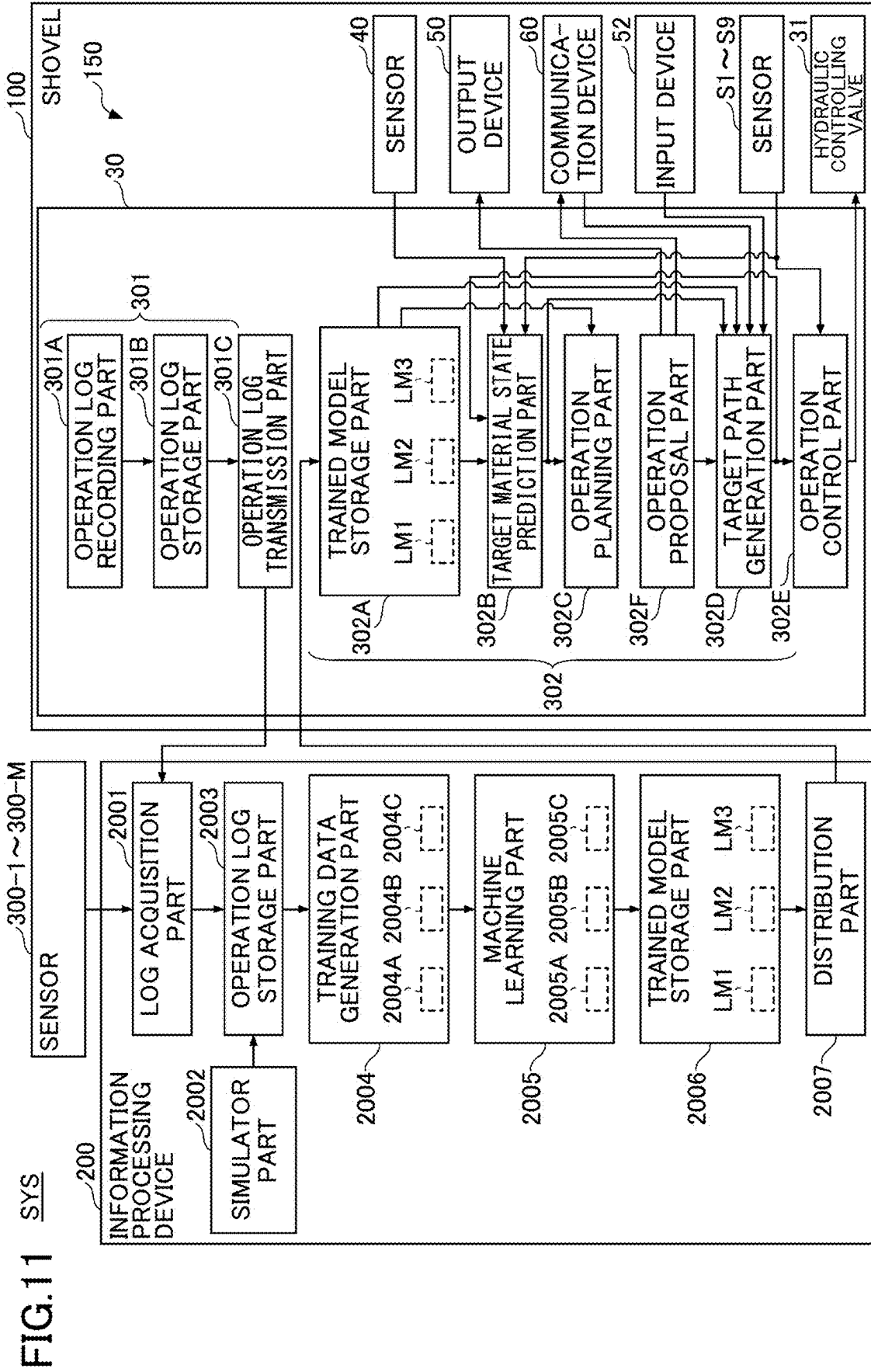


FIG.12

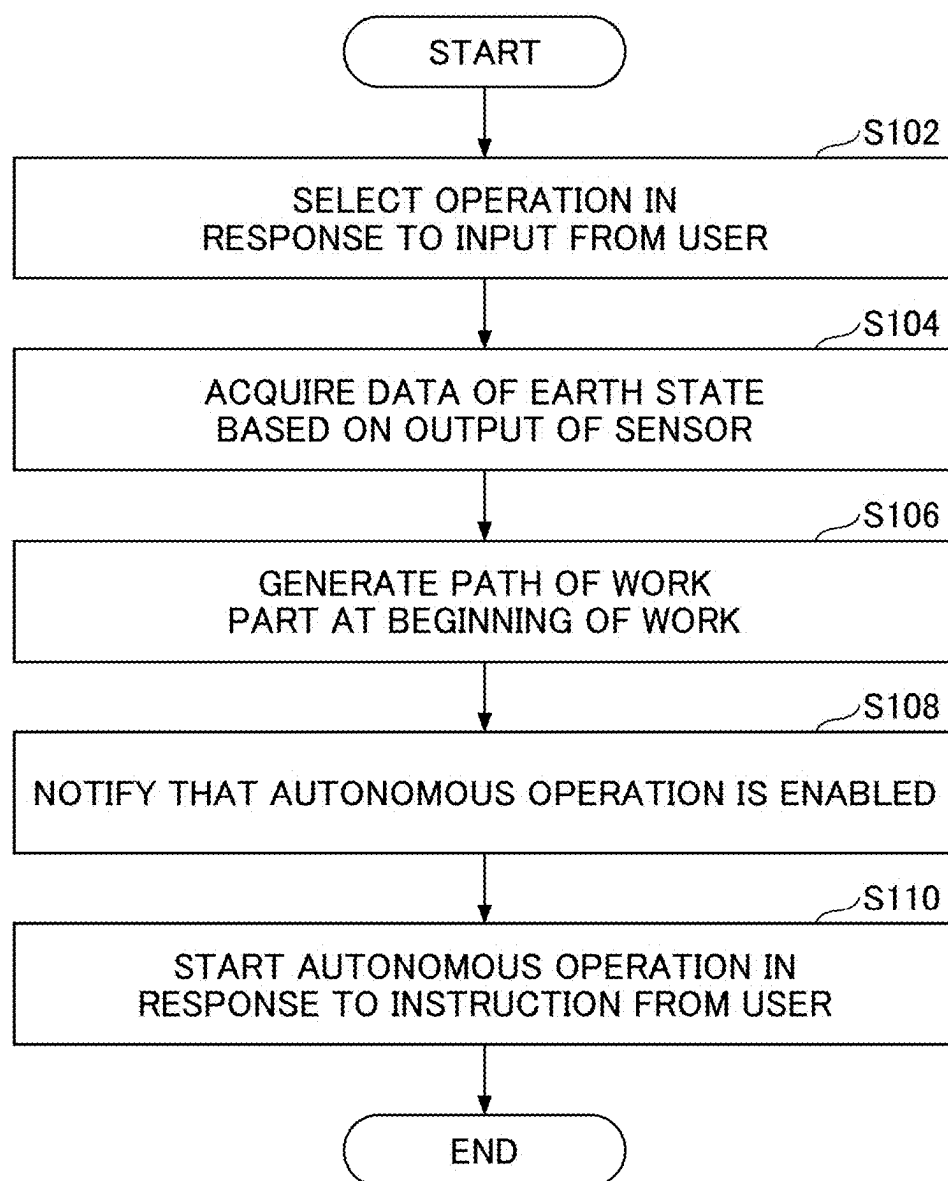


FIG.13

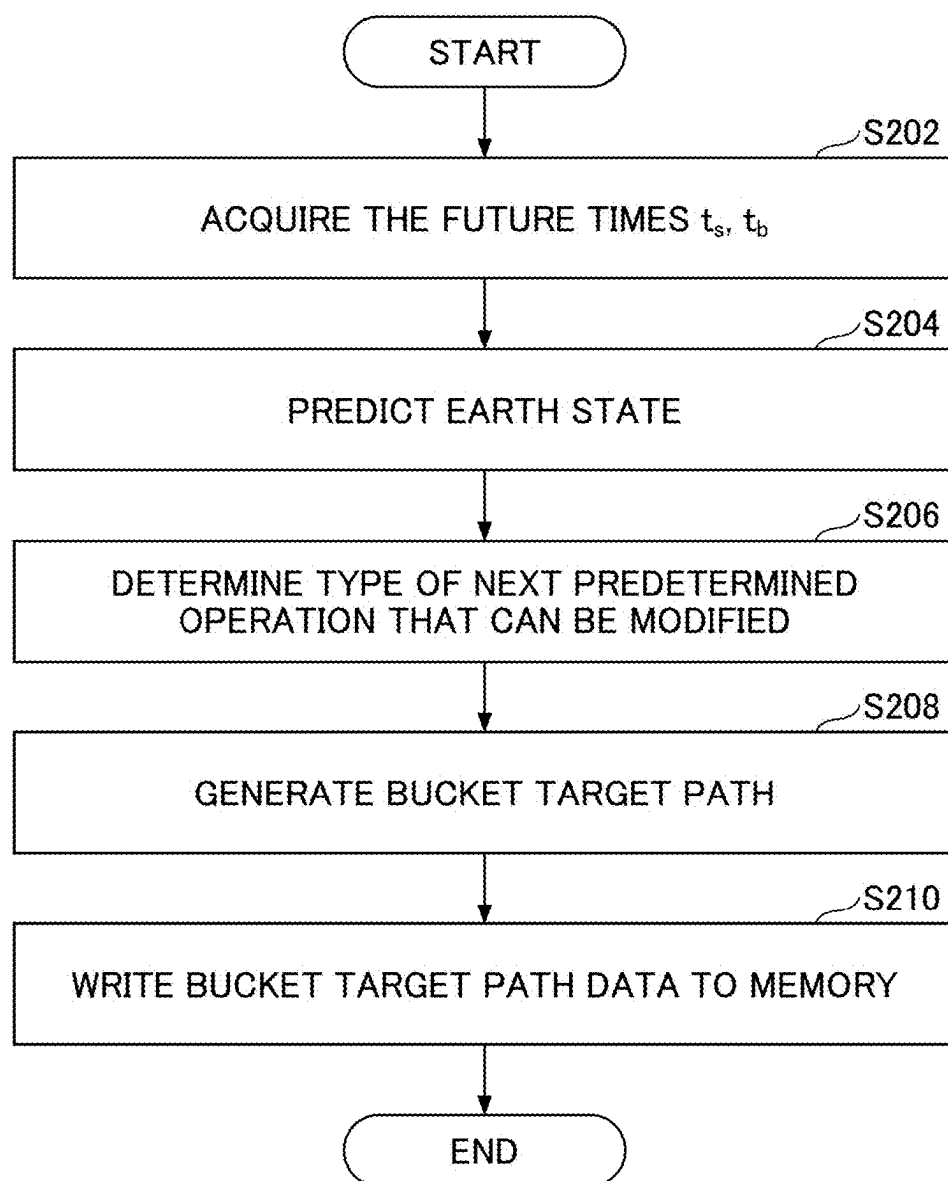


FIG.14

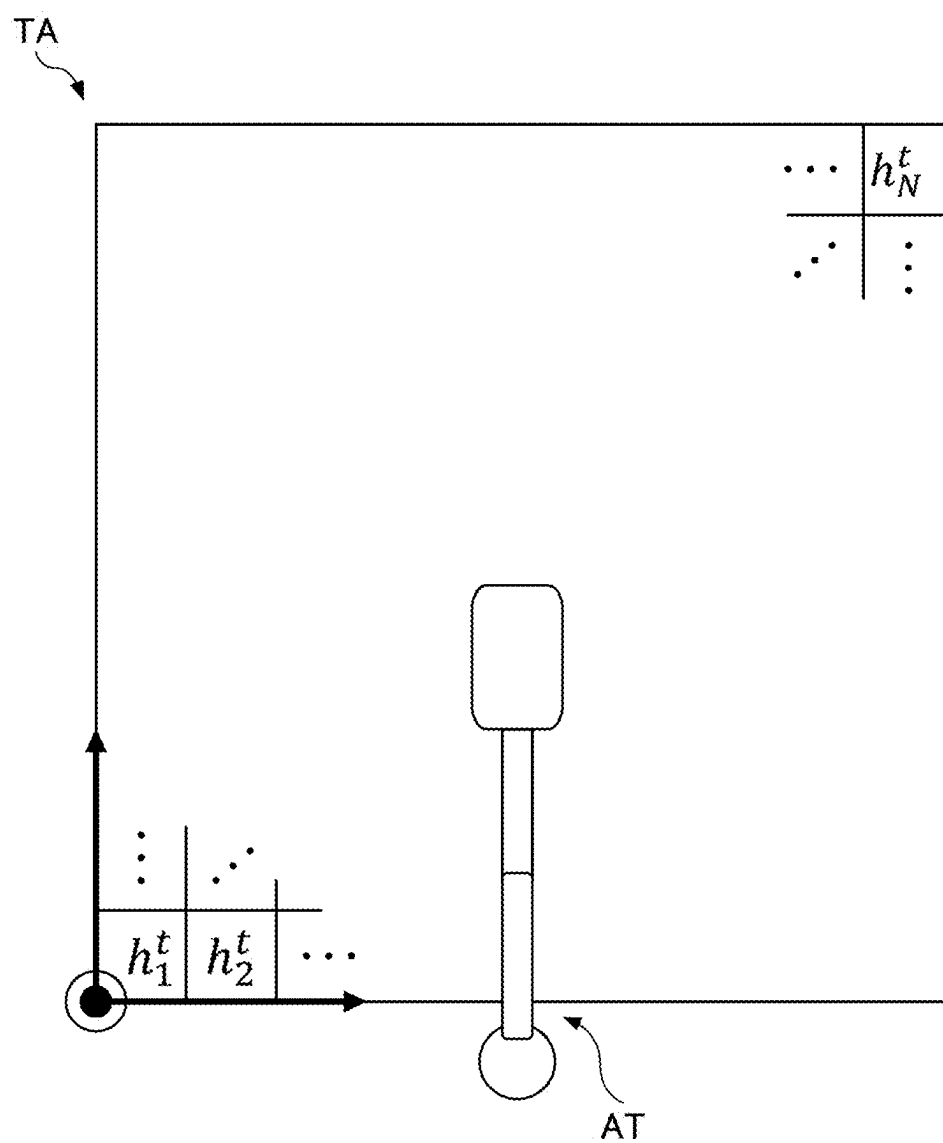


FIG.15

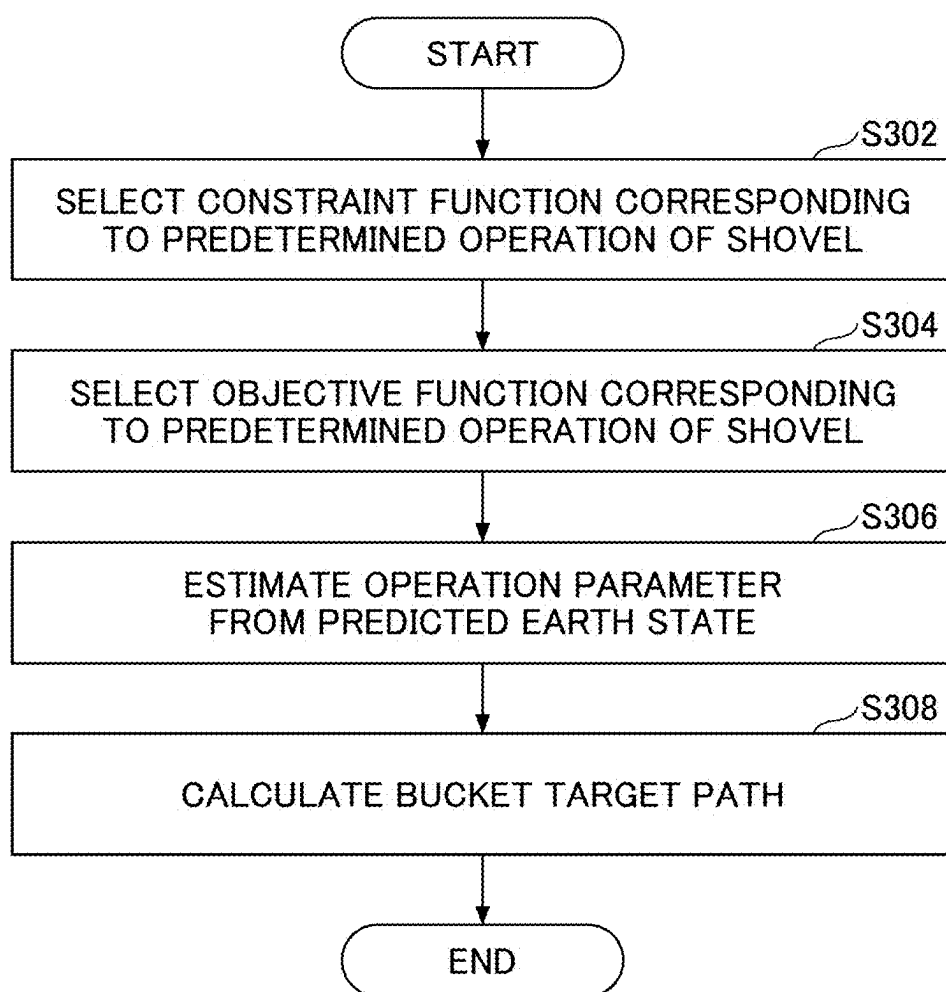


FIG.16

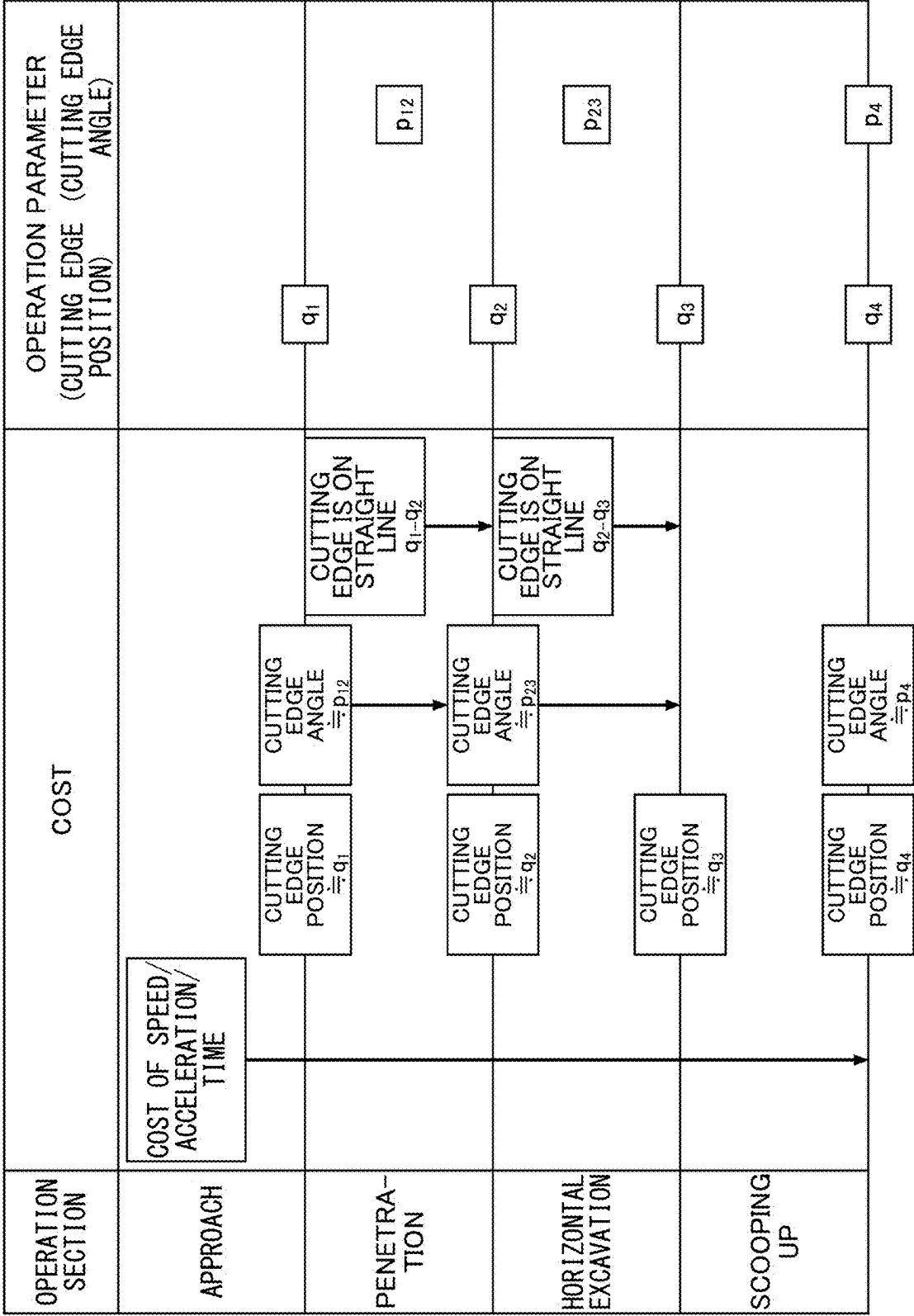
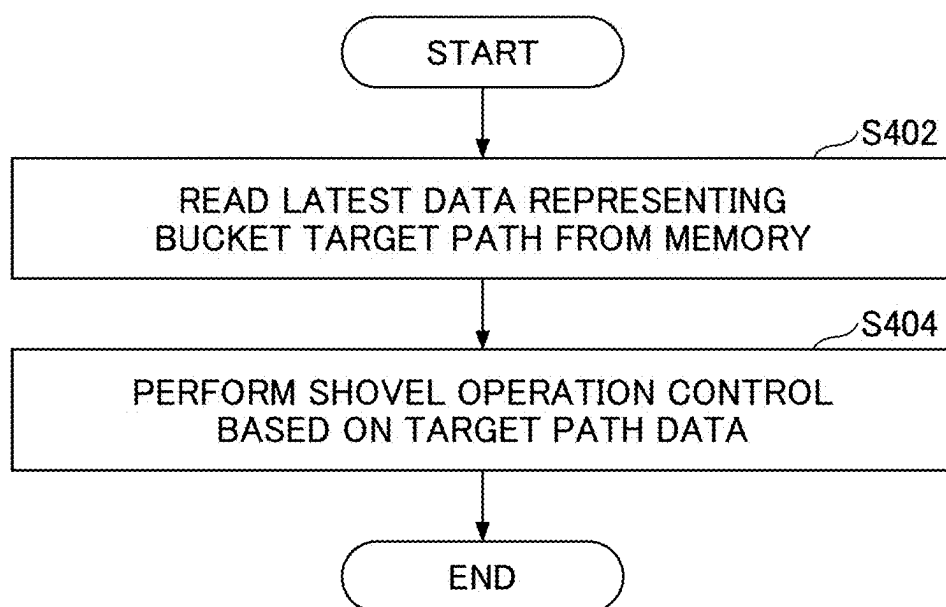


FIG.17



WORKING MACHINE, INFORMATION PROCESSING DEVICE, AND PROGRAM

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation application of International Application No. PCT/JP2023/041864, filed on Nov. 21, 2023, and designated the U.S., which is based upon and claims priority to Japanese Patent Application no. 2022-186780, filed on Nov. 22, 2022. The entire contents of these applications are incorporated herein by reference.

BACKGROUND

[0002] The disclosures herein relate to working machines, and the like.

[0003] Working machines such as shovels are known in a related art.

SUMMARY

[0004] According to one embodiment of the present disclosure, an information processing device includes an operation planning part configured to determine a future operation of a working machine from among a plurality of operations according to a performance status of an operation of the working machine.

[0005] According to another embodiment of the present disclosure, a non-transitory computer-readable recording medium having a program embodied therein causes an information processing device to determine a future operation of a working machine from among a plurality of operations according to a performance status of an operation of the working machine.

[0006] According to further another embodiment of the present disclosure, a non-transitory computer-readable recording medium having a program embodied therein a support to perform causes device determining a future operation of a working machine from among a plurality of operations according to a performance status of an operation of the working machine, and notifying an operator of the determined future operation.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a drawing illustrating an example of an operation support system;

[0008] FIG. 2 is a top view illustrating an example of a shovel;

[0009] FIG. 3 is a drawing illustrating an example of a configuration related to remote control of the shovel;

[0010] FIG. 4 is a drawing illustrating an example of a hardware configuration of the shovel;

[0011] FIG. 5 is a drawing illustrating an example of a hardware configuration of an information processing device;

[0012] FIG. 6 is a functional block diagram illustrating a first example of a functional configuration of the operation support system;

[0013] FIG. 7 is a drawing describing an example of a relationship between a timing of processing in shovel operation planning and a planned operation;

[0014] FIG. 8 is a drawing describing another example of the relationship between the timing of processing in shovel operation planning and the planned operation;

[0015] FIG. 9 is a state transition diagram illustrating an example of a transition of a shovel operation in slope construction work;

[0016] FIG. 10 is a state transition diagram illustrating an example of a transition of a shovel operation in ground leveling work;

[0017] FIG. 11 is a functional block diagram illustrating a second example of the functional configuration of the operation support system;

[0018] FIG. 12 is a flowchart schematically illustrating an example of processing related to a start of an autonomous operation of the shovel;

[0019] FIG. 13 is a main flowchart schematically illustrating an example of processing related to shovel operation planning and bucket path generation;

[0020] FIG. 14 is a drawing illustrating an example of an observation target area;

[0021] FIG. 15 is a sub-flowchart schematically illustrating an example of processing related to the bucket path generation;

[0022] FIG. 16 is a drawing illustrating an example of cost conditions and operation parameters corresponding to a plurality of sections of a shovel excavation operation; and

[0023] FIG. 17 is a flowchart schematically illustrating an example of processing related to shovel operation control.

DETAILED DESCRIPTION

[0024] It is desirable for a working machine to perform a more appropriate operation according to various conditions from the viewpoint of work efficiency, for example.

[0025] Therefore, in view of the above problems, it is an object to provide a technology that enables the working machine to perform the more appropriate operation.

[0026] According to the embodiments of the present disclosure, the working machine can perform the more appropriate operation.

[0027] In the following, embodiments of the present disclosure will be described with reference to the accompanying drawings.

[Outline of Operation Support System]

[0028] An outline of an operation support system SYS according to the present embodiment will be described with reference to FIGS. 1 to 3.

[0029] FIG. 1 is a drawing illustrating an example of the operation support system SYS. In FIG. 1, a left side view of a shovel 100 is shown. FIG. 2 is a top view illustrating an example of the shovel 100. FIG. 3 is a drawing illustrating an example of a configuration related to remote control of the shovel 100. Hereinafter, a direction in the shovel 100, or a direction viewed from the shovel 100 may be described by defining a direction in which an attachment AT extends in the top view of the shovel 100 (upward direction in FIG. 2) as “front”.

[0030] As shown in FIG. 1, the operation support system SYS includes the shovel 100, an information processing device 200, and a sensor group 300.

[0031] The operation support system SYS supports the operation of the shovel 100 by coordinating with the shovel 100 using the information processing device 200.

[0032] The shovel 100 included in the operation support system SYS may be one or a plurality of shovels.

[0033] The shovel 100 is a working machine to be supported in the operation support system SYS.

[0034] As shown in FIGS. 1 and 2, the shovel 100 includes a lower traveling body 1, an upper swivel body 3, the attachment AT including a boom 4, an arm 5, and a bucket 6, and a cabin 10.

[0035] The lower traveling body 1 uses crawlers 1C to cause the shovel 100 to travel. The crawlers 1C include a left crawler 1CL and a right crawler 1CR. The crawler 1CL is hydraulically driven by a traveling hydraulic motor 1ML. Similarly, the crawler 1CR is hydraulically driven by a traveling hydraulic motor 1MR. Thus, the lower traveling body 1 can move by itself.

[0036] The upper swivel body 3 is mounted on the lower traveling body 1 via a swivel mechanism 2 so as to be capable of swiveling (freely swivel). For example, the upper swivel body 3 swivels with respect to the lower traveling body 1 by hydraulically driving the swivel mechanism 2 by a swivel hydraulic motor 2M.

[0037] The boom 4 is attached to the front center of the upper swivel body 3 so that it is capable of lifting with respect to a rotation axis along the lateral direction. The arm 5 is rotatably attached to the tip of the boom 4 around a rotation axis oriented laterally. The bucket 6 is rotatably attached to the tip of the arm 5 around a rotation axis oriented laterally.

[0038] The bucket 6 is an example of an end attachment, and is used, for example, for excavation work, slope work, and ground leveling work.

[0039] The bucket 6 is attached to the tip of the arm 5 in a manner that allows it to be easily replaced based on work requirements of the shovel 100. That is, in place of the bucket 6, a bucket of a different kind from the bucket 6, for example, a relatively large bucket, a bucket for slope work, a bucket for dredging work, or the like may be attached to the tip of the arm 5. An end attachment other than a bucket, for example, an agitator, a breaker, a crusher, or the like may be attached to the tip of the arm 5. A spare attachment, for example, a quick coupling or a tilt rotator, may be provided between the arm 5 and the end attachment.

[0040] The boom 4, the arm 5, and the bucket 6 are hydraulically driven by the boom cylinder 7, the arm cylinder 8, and the bucket cylinder 9, respectively.

[0041] The cabin 10 is a control cabin in which an operator rides and operates the shovel 100. The cabin 10 is mounted, for example, on the front left of the upper swivel body 3.

[0042] The shovel 100 includes as equipment a communication device 60 and can communicate with the information processing device 200 through a predetermined communication line NW.

[0043] The communication line NW includes, for example, a local area network (LAN) of a work site. The communication line NW may also include a wide area network (WAN). The wide area network includes, for example, a mobile communication network terminated at a base station, a satellite communication network using a communication satellite, and an Internet network. The communication line NW may also include, for example, a short-distance communication line based on a wireless communication standard such as WiFi or Bluetooth (registered trademark).

[0044] For example, the shovel 100 causes driven elements such as the lower traveling body 1 (i.e., a pair of right and left crawlers 1CL and 1CR), the upper swivel body 3,

the boom 4, the arm 5, and the bucket 6 to operate according to the operation of the operator in the cabin 10.

[0045] In addition, the shovel 100 may be configured to be operable by the operator in the cabin 10, or may be configured to be remotely operated (remote control) from the outside of the shovel 100. When the shovel 100 is remotely operated, the interior of the cabin 10 may be unmanned. When the shovel 100 is dedicated to remote control, the cabin 10 may be omitted. Hereinafter, the description proceeds on the assumption that the operation of the operator includes at least one of the operation of an operating device 26 of the operator in the cabin 10 and the remote control of an external operator.

[0046] For example, as shown in FIG. 3, the remote control includes a mode in which the shovel 100 is operated by the operation input related to an actuator of the shovel 100 performed by the remote control support device 400 which can communicate with the shovel 100 through the communication line NW. The remote control support device 400 may be provided separately from the information processing device 200 or may be the information processing device 200.

[0047] The remote control support device 400 may be provided, for example, in a management center or the like which manages the work of the shovel 100 from the outside. Further, the remote control support device 400 may be a portable operation terminal, and in this case, the operator can perform the remote control of the shovel 100 while directly confirming the work status of the shovel 100 from surroundings of the shovel 100.

[0048] The shovel 100 may transmit an image (hereinafter, “surrounding image”) representing the surrounding state including the front of the shovel 100 based on the captured image output by an imaging device mounted in the shovel to the remote control support device 400 through, for example, the communication device 60 described later. Further, the shovel 100 may transmit the captured image output from the imaging device to the remote control support device 400 through the communication device 60, and the remote control support device 400 may process the captured image received from the shovel 100 to generate the surrounding image. Then, the remote control support device 400 may display the surrounding images representing the surrounding condition including the front of the shovel 100 on its own display device. In addition, various information images (information screens) displayed on an output device 50 (display device) inside the cabin 10 of the shovel 100 may similarly be displayed on the display device of the remote control support device 400. Thus, the operator using the remote control support device 400 can remotely operate the shovel 100 while confirming, for example, the display contents of the information screen and the image representing the surrounding condition of the shovel 100 displayed on the display device. Then, the shovel 100 may operate the actuator to drive the driven elements such as the lower traveling body 1, the upper swivel body 3, the boom 4, the arm 5, and the bucket 6 according to the remote control signal representing the contents of the remote control received from the remote control support device 400 by the communication device 60.

[0049] Further, the remote control may include a mode in which the shovel 100 is operated by, for example, external voice input or gesture input to the shovel 100 by a person (e.g., worker) around the shovel 100. More specifically, the

shovel 100 recognizes voices spoken by surrounding workers, gestures performed by workers, or the like, through a voice input device (e.g., microphone), a gesture input device (e.g., imaging device), or the like mounted in the shovel. Then, the shovel 100 may operate the actuator to drive the driven elements such as the lower traveling body 1 (right and left crawlers 1C), the upper swivel body 3, the boom 4, the arm 5, and the bucket 6 according to the contents of the recognized voices, gestures, or the like.

[0050] The shovel 100 may automatically operate the actuator regardless of the contents of the operator's operation. Thus, the shovel 100 can achieve a function of automatically operating at least a part of the driven elements such as the lower traveling body 1, the upper swivel body 3, and the attachment AT, that is, what is called an "automatic operation function" or a "Machine Control (MC) function".

[0051] The automatic operation function includes, for example, a semi-automatic operation function (operation-assisted MC function). The semi-automatic operation function is a function of automatically operating driven elements (actuators) other than the driven element (actuator) to be operated in response to the operator's operation. The automatic operation function may also include a fully automatic operation function (fully automatic MC function). The fully automatic operation function is a function of automatically operating at least a part of the plurality of driven elements (hydraulic actuators) on the assumption that there is no operator's operation. When the fully automatic operation function is enabled in the shovel 100, the cabin 10 may be in an unmanned state. When the shovel 100 is dedicated to fully automatic operation, the cabin 10 may be omitted. The semi-automatic operation function and the fully automatic operation function include, for example, a rule-based automatic operation function. The rule-based automatic operation function is an automatic operation function in which the operation contents of the driven element (actuator) to be operated automatically are determined according to a previously defined rule. The semi-automatic operation function and the fully automatic operation function may include an autonomous operation function. The autonomous operation function is an automatic operation function in which the shovel 100 autonomously makes various determinations, and the operation contents of the driven element (hydraulic actuator) to be operated automatically are determined according to the determination results.

[0052] In addition, the operation of the shovel 100 may be remotely monitored. In this case, a remote monitoring support device having the same function as the remote control support device 400 may be provided. The remote monitoring support device is, for example, an information processing device 200. Thus, a monitor who is a user of the remote monitoring support device can monitor the operation of the shovel 100 while checking the surrounding image displayed on the display device of the remote monitoring support device. In addition, if the monitor determines that it is necessary from the viewpoint of safety, for example, the monitor can intervene in the operation or automatic operation of the shovel 100 by using the input device of the remote monitoring support device and by making a predetermined input, and can make the shovel 100 stop urgently.

[0053] The information processing device 200 coordinates with the shovel 100 by communicating with each other and supports the operation of the shovel 100.

[0054] The information processing device 200 is, for example, a server device or a terminal device for management installed in a management office in the work site of the shovel 100, or in a management center or the like that manages the operation status of the shovel 100 at a place different from the work site of the shovel 100. The server device may be an on-premises server, a cloud server, or an edge server. The terminal device for management may be, for example, a stationary terminal device such as a desktop PC (Personal Computer), or a portable terminal device (portable terminal) such as a tablet terminal, a smartphone, or a laptop PC. In the latter case, a worker at the work site, a supervisor who supervises the work, or a manager who manages the work site can carry the portable information processing device 200 and move around the work site. In the latter case, an operator can carry the portable information processing device 200 into the cabin of the shovel 100, for example.

[0055] The information processing device 200 acquires data related to the operating state from the shovel 100, for example. Thus, the information processing device 200 can determine the operating state of the shovel 100 and monitor presence or absence of abnormality of the shovel 100. In addition, the information processing device 200 can display data related to the operating state of the shovel 100 through the display device 208 described later, for example, and allow the user to confirm the data. The information processing device 200 can, for example, make a learning model learn the operating state of the shovel 100 and generate a trained model for supporting the operation of the shovel 100.

[0056] In addition, the information processing device 200 may transmit to the shovel 100 various data such as programs and reference data used in the processing of a controller 30 or the like to the shovel 100. Thus, the shovel 100 can perform various processes related to the operation of the shovel 100 using various data downloaded from the information processing device 200.

[0057] The sensor group 300 is installed at the work site of the shovel 100. Target material includes, for example, earth in the work area around the shovel 100.

[0058] For example, when a plurality of shovels 100 are included in the operation support system SYS, the sensor group 300 is provided for each shovel 100. In addition, when a plurality of shovels 100 included in the operation support system SYS perform work at the same work site, one sensor group 300 may be shared for the plurality of shovels 100.

[0059] The sensor group 300 includes sensors 300-1 to 300-M (M: an integer of 2 or more). The sensors 300-1 to 300-M measure the state of objects in the work site around the shovel 100 and acquire measurement data related to the state. The objects in the work site include, for example, target materials (earth in the work area) around the shovel 100, working machines such as other shovels and bulldozers, and working vehicles such as trucks for transporting earth around the shovel 100. The state of the objects includes a shape and characteristic of the objects.

[0060] The sensors 300-1 to 300-M include, for example, ranging sensors (distance sensors). The ranging sensors include, for example, LIDAR (Light Detecting and Ranging), millimeter wave radar, ultrasonic sensors, infrared sensors, and the like. The sensors 300-1 to 300-M may also include, for example, a stereo camera, a TOF (Time of Flight) camera, and a 3D camera capable of acquiring data on distance (depth) in addition to two-dimensional images.

The sensors **300-1** to **300-M** may also include both the ranging sensor and the 3D camera. Thus, the sensor group **300** can acquire measurement data representing the shape of the object at the work site around the shovel **100**. Hereinafter, a sensor capable of acquiring measurement data representing the shape of the object, such as a ranging sensor or a 3D camera, may be conveniently referred to as a “shape sensor”.

[0061] In addition, the sensors **300-1** to **300-M** may include a multi-wavelength spectral camera. The multi-wavelength spectral camera includes, for example, a multi-spectral camera or a hyperspectral camera. Thus, for example, the sensor group **300** can acquire measurement data representing a characteristic of the object at the work site around the shovel **100**, such as hardness and moisture content of the earth. Hereinafter, a sensor capable of acquiring measurement data representing the characteristics of the object, such as a multi-wavelength spectral camera, may be conveniently referred to as a “characteristic sensor”.

[0062] For example, the sensors **300-1** to **300-M** include a plurality of shape sensors. The plurality of shape sensors may be provided at different locations in the work site around the shovel **100**, with their sensing ranges overlapping at least one other shape sensor's sensing range. Thus, for example, even if one shape sensor's measurement data cannot acquire data representing the shape of a part of an object within the sensing range due to occlusion, other shape sensors may be able to acquire the measurement data for the part of the object. Therefore, the sensor group **300** can more reliably acquire measurement data representing the shape of an object in the work site around the shovel **100**.

[0063] Further, the sensors **300-1** to **300-M** may include a plurality of characteristic sensors. Further, the plurality of characteristic sensors may be provided at different locations in the work site around the shovel **100**, with their sensing ranges overlapping at least one other characteristic sensor's sensing range. Thus, for example, even if one characteristic sensor's measurement data cannot acquire data representing the shape of a part of an object within the sensing range due to occlusion, other characteristic sensors may be able to acquire the measurement data for the part of the object. Therefore, the sensor group **300** can more reliably acquire measurement data representing the characteristics of an object in the work site around the shovel **100**.

[0064] In addition, the sensors **300-1** to **300-M** may include a sensor having both the function of a shape sensor and the function of a characteristic sensor. In this case, the sensors **300-1** to **300-M** may include a plurality of integrated sensors (hereinafter, “integrated sensor”). Further, the plurality of integrated sensors may be provided at different locations in the work site around the shovel **100**, with their sensing ranges overlapping at least one other integrated sensor's sensing range.

[0065] The sensor group **300** may simply include only one shape sensor or characteristic sensor. In place of the sensor group **300**, the operation support system **SYS** may simply include only one sensor capable of acquiring measurement data related to the state of an object in the work site around the shovel **100**.

[0066] The sensors **300-1** to **300-M** may be fixed to the work site around the shovel **100** or mounted on a mobile body movable in the work site around the shovel **100**. The mobile body may include, for example, a working machine or a working vehicle moving in the work site. The mobile

body movable in the work site may include, for example, a flying body such as a drone flying over the work site.

[0067] The output (measured data) of the sensors **300-1** to **300-M** is received into the information processing device **200** through the communication line **NW**. The output of the sensors **300-1** to **300-M** is received directly into the information processing device **200** through the communication line **NW**, for example. The output of the sensors **300-1** to **300-M** may be received into the shovel **100** through the communication line **NW** and received into the information processing device **200** via the shovel **100**. When the sensors **300-1** to **300-M** are mounted on a predetermined device such as the above-described mobile device, the output of the sensors **300-1** to **300-M** may be received into the predetermined device and received into the information processing device **200** from the device.

[Hardware Configuration of Operation Support System]

[0068] Next, the hardware configuration of the operation support system **SYS** will be described with reference to FIGS. **4** and **5** in addition to FIGS. **1** to **3**.

[0069] The hardware configuration of the remote control support device **400** may be the same as that of the information processing device **200**. Therefore, illustration and description of the hardware configuration of the remote control support device **400** will be omitted.

<Hardware Configuration of Shovel>

[0070] FIG. **4** is a block diagram illustrating an example of a hardware configuration of a shovel **100**.

[0071] In FIG. **4**, a path through which mechanical power is transmitted is indicated by a double line, paths through which high-pressure hydraulic fluid for driving the hydraulic actuators flows are indicated by solid lines, paths through which pilot pressure is transmitted are indicated by broken lines, and paths through which electrical signals are transmitted are indicated by dotted lines.

[0072] The shovel **100** includes components such as a hydraulic drive system for hydraulically driving the driven element, an operation system for operating the driven element, user interface system for exchanging information with the user, a communication system for communicating with the outside, and a control system for various controls.

«Hydraulic Drive System»

[0073] As shown in FIG. **4**, the hydraulic drive system of the shovel **100** includes hydraulic actuators **HA** for hydraulically driving each of the driven elements such as the lower traveling body **1** (left and right crawlers **1C**), the upper swivel body **3**, the boom **4**, the arm **5**, and the bucket **6**, as described above. The hydraulic drive system of the shovel **100** according to the present embodiment includes an engine **11**, a regulator **13**, a main pump **14**, and a control valve **17**.

[0074] The hydraulic actuators **HA** include such as traveling hydraulic motors **1ML** and **1MR**, a swivel hydraulic motor **2M**, a boom cylinder **7**, an arm cylinder **8**, and a bucket cylinder **9**.

[0075] In the shovel **100**, some or all of the hydraulic actuators **HA** may be replaced with an electric actuator. In other words, the shovel **100** may be a hybrid shovel or an electric shovel.

[0076] The engine **11** is a motor of the shovel **100** and a main power source in the hydraulic drive system. The engine

11 is, for example, a diesel engine using light oil as fuel. The engine **11** is mounted, for example, at the rear of the upper swivel body **3**. The engine **11** rotates at a predetermined target speed, for example, directly or indirectly controlled by the controller **30** to be described later, and drives the main pump **14** and a pilot pump **15**.

[0077] In place of or in addition to the engine **11**, another motor (e.g., electric motor) or the like may be mounted in the shovel **100**.

[0078] The regulator **13** controls (adjusts) a discharge amount of the main pump **14** under the control of the controller **30**. For example, the regulator **13** adjusts an angle (hereinafter, “tilt angle”) of a swash plate of the main pump **14** according to a control command from the controller **30**.

[0079] The main pump **14** supplies hydraulic fluid to the control valve **17** through a high-pressure hydraulic line. The main pump **14** is mounted, for example, at the rear of the upper swivel body **3** as well as the engine **11**. The main pump **14** is driven by the engine **11** as described above. The main pump **14** is, for example, a variable displacement hydraulic pump, and the stroke length of a piston is adjusted by adjusting the tilt angle of the swash plate by the regulator **13** under the control of the controller **30**, and the discharge flow rate and discharge pressure are controlled as described above.

[0080] The control valve **17** drives the hydraulic actuators HA in accordance with the operator’s operation or remote control of the operating device **26** or an operation command corresponding to the automatic operation function. The control valve **17** is mounted, for example, in the center of the upper swivel body **3**. The control valve **17** is connected to the main pump **14** via a high-pressure hydraulic line as described above, and selectively supplies hydraulic oil supplied from the main pump **14** to the respective hydraulic actuators HA in accordance with the operator’s operation or an operation command corresponding to the automatic operation function. Specifically, the control valve **17** includes a plurality of controlling valves (direction switching valves) for controlling the flow rate and direction of the hydraulic oil supplied from the main pump **14** to the hydraulic actuators HA.

«Operation System»

[0081] As shown in FIG. 4, the operation system of the shovel **100** includes a pilot pump **15**, an operating device **26**, a hydraulic controlling valve **31**, a shuttle valve **32**, and a hydraulic controlling valve **33**.

[0082] The pilot pump **15** supplies the pilot pressure to various hydraulic devices via a pilot line **25**. The pilot pump **15** is mounted, for example, in the rear part of the upper swivel body **3**, similar to the engine **11**. The pilot pump **15** is, for example, a fixed displacement hydraulic pump, and is driven by the engine **11** as described above.

[0083] The pilot pump **15** may be omitted. In this case, after the relatively high pressure hydraulic fluid discharged from the main pump **14** is depressurized by a predetermined pressure reducing valve, the relatively low pressure hydraulic fluid may be supplied to various hydraulic devices as the pilot pressure.

[0084] The operating device **26** is provided near an operator’s seat of the cabin **10**, and is used for an operator to operate various driven elements. Specifically, the operating device **26** is used for an operator to operate the hydraulic actuators HA that drives each driven element, and as a result,

the operator can operate the driven element to be driven by the hydraulic actuators HA. The operating device **26** includes a pedal device and a lever device for operating each driven element (hydraulic actuator HA).

[0085] For example, as shown in FIG. 4, the operating device **26** is a hydraulic pilot type. Specifically, the operating device **26** uses hydraulic fluid supplied from the pilot pump **15** through the pilot line **25** and a branched pilot line **25A**, and outputs a pilot pressure corresponding to the operation contents to a secondary pilot line **27A**. The pilot line **27A** is connected to one inlet port of the shuttle valve **32**, and is connected to the control valve **17** through a pilot line **27** connected to an outlet port of the shuttle valve **32**. Thus, the pilot pressure corresponding to the operation contents of various driven elements (hydraulic actuators HA) in the operating device **26** can be input to the control valve **17** through the shuttle valve **32**. Therefore, the control valve **17** can drive the hydraulic actuators HA according to the operation contents of the operating device **26** by the operator or the like.

[0086] The operating device **26** may be an electric type. In this case, the pilot line **27A**, the shuttle valve **32**, and the hydraulic controlling valve **33** are omitted. Specifically, the operating device **26** outputs an electric signal (hereinafter, “operation signal”) according to the operation contents, and the operation signal is received into the controller **30**. Then, the controller **30** outputs a control command according to the operation contents, that is, a control signal according to the operation contents for the operating device **26**, to the hydraulic controlling valve **31**. Thus, the pilot pressure corresponding to the operation contents of the operating device **26** are input from the hydraulic controlling valve **31** to the control valve **17**, and the control valve **17** can drive the respective hydraulic actuators HA according to the operation contents of the operating device **26**.

[0087] The controlling valve (direction switching valve) built in the control valve **17** and driving the respective hydraulic actuators HA may be an electromagnetic solenoid type. In this case, the operation signal output from the operating device **26** may be directly input to the control valve **17** (i.e., to the electromagnetic solenoid type controlling valve).

[0088] In addition, as described above, some or all of the hydraulic actuators HA may be replaced with electric actuators. In this case, the controller **30** may output a control command corresponding to the operation content of the operating device **26** or the remote control content defined by the remote control signal to the electric actuator or a driver for driving the electric actuator. When the shovel **100** is operated remotely, the operating device **26** may be omitted.

[0089] The hydraulic controlling valve **31** is provided for each driven element (hydraulic actuator HA) to be operated by the operating device **26** and for each driving direction (e.g., the upward and downward directions of the boom **4**) of the driven element (hydraulic actuator HA). For example, two hydraulic controlling valves **31** are provided for each double-acting hydraulic actuator HA for driving the lower traveling body **1**, the upper swivel body **3**, the boom **4**, the arm **5**, and the bucket **6**, and the like. The hydraulic controlling valve **31** may be provided, for example, in the pilot line **25B** between the pilot pump **15** and the control valve **17**, and may be configured so that the flow path area (i.e., the cross-sectional area through which the hydraulic fluid can flow) thereof can be changed. Thus, the hydraulic

controlling valve **31** can output a predetermined pilot pressure to a secondary pilot line **27B** by utilizing the hydraulic fluid of the pilot pump **15** supplied through the pilot line **25B**. Therefore, the hydraulic controlling valve **31** can indirectly apply a predetermined pilot pressure corresponding to a control signal from the controller **30** to the control valve **17** through the shuttle valve **32** between the pilot line **27B** and the pilot line **27**. Thus, for example, the controller **30** can cause the hydraulic controlling valve **31** to supply the pilot pressure corresponding to an operation command corresponding to the automatic operation function to the control valve **17** to achieve the operation of the shovel **100** by the automatic operation function.

[0090] Further, the controller **30** may control the hydraulic controlling valve **31** to achieve the remote control of the shovel **100**. Specifically, the controller **30** outputs to the hydraulic controlling valve **31** a control signal corresponding to the content of the remote control specified by the remote control signal received from the remote control support device **400** by the communication device **60**. Thus, the controller **30** can cause the hydraulic controlling valve **31** to supply the pilot pressure corresponding to the content of the remote control to the control valve **17** to achieve the operation of the shovel **100** based on the remote control of the operator.

[0091] When the operating device **26** is an electric type, the controller **30** directly supplies the pilot pressure corresponding to the operation content (operation signal) of the operating device **26** from the hydraulic controlling valve **31** to the control valve **17** to achieve the operation of the shovel **100** based on the operation of the operator.

[0092] The shuttle valve **32** has two inlet ports and one outlet port, and outputs hydraulic fluid having the higher pilot pressure among the pilot pressures input to the two inlet ports to the outlet port. Like the hydraulic controlling valve **31**, the shuttle valve **32** is provided for each driven element (hydraulic actuator HA) to be operated by the operating device **26** and for each driving direction of the driven element (hydraulic actuator HA). For example, two shuttle valves **32** are provided for each double-acting hydraulic actuator HA for driving the lower traveling body **1**, the upper swivel body **3**, the boom **4**, the arm **5**, and the bucket **6**. One of the two inlet ports of the shuttle valve **32** is connected to the secondary pilot line **27A** of the operating device **26** (specifically, the above-described lever device and pedal device included in the operating device **26**), and the other is connected to the secondary pilot line **27B** of the hydraulic controlling valve **31**. The outlet port of the shuttle valve **32** is connected to the pilot port of the corresponding control valve of the control valve **17** through the pilot line **27**. The corresponding control valve is a control valve for driving the hydraulic actuator HA to be operated by the above-described lever device or pedal device connected to one inlet port of the shuttle valve **32**. Therefore, these shuttle valves **32** can apply the higher of the pilot pressure of the secondary pilot line **27A** of the operating device **26** and the pilot pressure of the secondary pilot line **27B** of the hydraulic controlling valve **31** to the pilot port of the corresponding control valve. In other words, the controller **30** can control the corresponding control valve by outputting the pilot pressure higher than the pilot pressure of the secondary side of the operating device **26** from the hydraulic controlling valve **31** regardless of the operator's operation of the operating device **26**. Therefore, the controller **30** can control the operation of the

driven elements (lower traveling body **1**, upper swivel body **3**, boom **4**, arm **5**, and bucket **6**) and achieve the automatic operation function and the remote control function regardless of the operation state of the operator with respect to the operating device **26**.

[0093] The hydraulic controlling valve **33** is provided in the pilot line **27A** connecting the operating device **26** and the shuttle valve **32**. The hydraulic controlling valve **33** is configured so that the flow path area can be changed, for example. The hydraulic controlling valve **33** operates according to a control signal input from the controller **30**. Thus, when the operating device **26** is operated by an operator, the controller **30** can forcibly reduce the pilot pressure output from the operating device **26**. Therefore, even when the operating device **26** is operated, the controller **30** can forcibly reduce or stop the operation of the hydraulic actuator HA corresponding to the operation of the operating device **26**. Moreover, even when the operating device **26** is operated, for example, the controller **30** can reduce the pilot pressure output from the operating device **26** to be lower than the pilot pressure output from the hydraulic controlling valve **31**. Therefore, by controlling the hydraulic controlling valve **31** and the hydraulic controlling valve **33**, the controller **30** can surely cause the desired pilot pressure to act on the pilot port of the control valve in the control valve **17** regardless of the operation content of the operating device **26**, for example. Therefore, by controlling the hydraulic controlling valve **33** in addition to the hydraulic controlling valve **31**, the controller **30** can more appropriately achieve the automatic operation function and the remote control function of the shovel **100**.

«User Interface System»

[0094] As shown in FIG. 4, the user interface system of the shovel **100** includes an operating device **26**, an output device **50**, and an input device **52**.

[0095] The output device **50** outputs various types of information to a user of the shovel **100** (e.g., the operator in the cabin **10** or the external operator of the remote control) or to a person around the shovel **100** (e.g., a worker or a driver of a working vehicle). For example, the output device **50** includes a lighting device and a display device for outputting various types of information in a visual manner. The lighting device is, for example, a warning lamp (indicator lamp). The display device is, for example, a liquid crystal or display an organic EL (Electroluminescence) display. For example, as shown in FIG. 2, the lighting device and the display device may be provided inside the cabin **10** and output various types of information in a visual manner to an operator or the like inside the cabin **10**. Further, the lighting apparatus and the display device may be provided, for example, on the lateral surface of the upper swivel body **3**, and may output various types of information in a visual manner to a worker or the like around the shovel **100**.

[0096] The output device **50** may also include a sound output device which outputs various types of information in an auditory method (see FIG. 7). The sound output device may include, for example, a buzzer or a speaker. The sound output device may be provided, for example, inside or outside the cabin **10**, and may output various types of information in an auditory method to an operator inside the cabin **10** or to a person (a worker, etc.) around the shovel **100**.

[0097] The output device **50** may also include a device which outputs various types of information in a tactile manner such as vibration of the operator's seat.

[0098] The input device **52** receives various types of inputs from the user of the shovel **100**, and signals corresponding to the received inputs are received into the controller **30**. For example, as shown in FIG. 2, the input device **52** is provided inside the cabin **10** and receives inputs from an operator inside the cabin **10**. The input device **52** may be provided, for example, on the lateral surface of the upper swivel body **3**, and may receive input from a worker or the like around the shovel **100**.

[0099] For example, the input device **52** may include an operation input device that accepts input by mechanical operation from a user. The operation input device may include a touch panel mounted on the display device, a touch pad installed around the display device, a button switch, a lever, a toggle, and a knob switch provided on the operating device **26** (lever device).

[0100] The input device **52** may also include a voice input device that accepts voice input from a user. The voice input device may include, for example, a microphone.

[0101] The input device **52** may also include a gesture input device that accepts gesture input from the user. The gesture input device may include, for example, an imaging device that captures a gesture performed by a user.

[0102] The input device **52** may also include a biometric input device that accepts biometric input of the user. The biometric input includes, for example, the input of biometric information as such the user's fingerprint or iris.

«Communication System»

[0103] As shown in FIG. 4, the communication system of the shovel **100** according to the present embodiment includes a communication device **60**.

[0104] The communication device **60** is connected to an external communication line NW and communicates with a device provided separately from the shovel **100**. The device provided separately from the shovel **100** may include a device outside the shovel **100** and a portable terminal device (portable terminal) brought into the cabin **10** by the user of the shovel **100**. The communication device **60** may include, for example, a mobile communication module conforming to a standard such as 4G (Fourth Generation) or 5G (Fifth Generation). The communication device **60** may include, for example, a satellite communication module. The communication device **60** may include, for example, a WiFi communication module or a Bluetooth (registered trademark) communication module. The communication device **60** may include a plurality of communication devices according to the type of communication line NW when there are a plurality of communication lines NW that can be connected.

[0105] For example, the communication device **60** communicates with external devices such as the information processing device **200** and the remote control support device **400** in the work site through a local communication line constructed at the work site. The local communication line may be, for example, a mobile communication line using local 5G (what is called local 5G) constructed at the work site or a local network using WiFi6.

[0106] The communication device **60** may communicate with the information processing device **200** and the remote control support device **400** located outside the work site

through a wide-area communication line including the work site, that is, a wide-area network.

«Control System»

[0107] As shown in FIG. 4, the control system of the shovel **100** includes a controller **30**. The control system of the shovel **100** according to the present embodiment includes an operating pressure sensor **29**, a sensor **40**, and sensors **S1** to **S9**.

[0108] The controller **30** performs various controls related to the shovel **100**.

[0109] Functions of the controller **30** may be achieved by any hardware or any combination of hardware and software. For example, as shown in FIG. 4, the controller **30** includes an auxiliary storage device **30A**, a memory device **30B**, a CPU (Central Processing Unit) **30C**, and an interface device **30D** connected by bus BS1.

[0110] The auxiliary storage device **30A** is a nonvolatile storage means and stores programs to be installed, and also stores necessary files, data, and the like. The auxiliary storage device **30A** may be, for example, an EEPROM (Electrically Erasable Programmable Read-Only Memory) or a flash memory.

[0111] The memory device **30B** loads the program of the auxiliary storage device **30A** so that the CPU **30C** can read it, for example, when a program start instruction is given. The memory device **30B** is, for example, an SRAM (Static Random Access Memory).

[0112] The CPU **30C** performs the program loaded into the memory device **30B**, for example, and implements various functions of the controller **30** according to the instruction of the program.

[0113] The interface device **30D** functions, for example, as a communication interface for connecting to the communication line inside the shovel **100**. The interface device **30D** may include a plurality of different types of communication interfaces according to the type of communication line to be connected.

[0114] The interface device **30D** also functions as an external interface for reading data from or writing data to the storage medium. The storage medium is, for example, a dedicated tool connected to a connector installed inside the cabin **10** by a detachable cable. The storage medium may be, for example, a general-purpose storage medium such as an SD memory card or a USB (Universal Serial Bus) memory. Thus, a program for realizing various functions of the controller **30** may be provided, for example, by a portable storage medium and installed in the auxiliary storage device **30A** of the controller **30**. The program may be downloaded from another computer (e.g., information processor **200**) outside the shovel **100** through the communication device **60** and installed in the auxiliary storage device **30A**.

[0115] It should be noted that a part of the functions of the controller **30** may be achieved by other controllers (control devices). In other words, the functions of the controller **30** may be achieved by a plurality of controllers installed in the shovel **100** in a distributed manner.

[0116] The operating pressure sensor **29** detects the pilot pressure on the secondary side (pilot line **27A**) of the hydraulic pilot type operating device **26**, that is, the pilot pressure corresponding to the operating state of each driven element (hydraulic actuator HA) in the operating device **26**. A detection signal of the pilot pressure corresponding to the operating state of each driven element (hydraulic actuator

HA) in the operating device 26 by the operating pressure sensor 29 is received into the controller 30.

[0117] When the operating device 26 is an electric type, the operating pressure sensor 29 is omitted. This is because the controller 30 can determine the operating state of each driven element through the operating device 26 based on the operating signal received from the operating device 26.

[0118] The sensor 40 acquires, for example, measurement data related to the shape of an object around the shovel 100.

[0119] For example, the sensor 40 is a shape sensor capable of acquiring measurement data representing the shape of an object around the shovel 100, such as a distance measurement sensor or a 3D camera. In addition to the function of the shape sensor, the sensor 40 may be an integrated sensor having the function of a characteristic sensor capable of acquiring measurement data representing the characteristic of an object around the shovel 100, such as a multi-wavelength spectral camera.

[0120] For example, as shown in FIG. 2, the sensor 40 includes sensors 40F, 40B, 40L, and 40R. The sensor 40F measures the state (shape and characteristics) of the object in front of the upper swivel body 3. The sensor 40B measures the state of the object in the upper swivel body 3. The sensor 40L measures the state of the object to the left of the upper swivel body 3. The sensor 40R measures the state of the object to the right of the upper swivel body 3. Thus, in the top view of the shovel 100, the sensor 40 can measure the state of the object in the whole circumference centering on the shovel 100, that is, in the angular direction of 360 degrees. Hereinafter, the sensors 40F, 40B, 40L, and 40R may be collectively or individually referred to as “sensor 40X”.

[0121] The output data (i.e., measurement data about the state of the object around the shovel 100) of the sensor 40 (sensor 40X) is received into the controller 30 through a one-to-one communication line or an in-vehicle network. Thus, for example, the controller 30 can determine the state of the shape and characteristics of the object around the shovel 100 based on the output data of the sensor 40X.

[0122] Some or all of the sensors 40B, 40L and 40R may be omitted.

[0123] The sensor S1 is attached to the boom 4 and measures an attitude state of the boom 4. The sensor S1 outputs measurement data indicating the attitude state of the boom 4. The attitude state of the boom 4 is, for example, the attitude angle (hereinafter, “boom angle”) around the rotation axis of the base end corresponding to a connecting part of the boom 4 with the upper swivel body 3. The sensor S1 includes, for example, a rotary potentiometer, a rotary encoder, an acceleration sensor, an angular acceleration sensor, a six-axis sensor, an IMU (Inertial Measurement Unit), and the like. Hereinafter, the same may apply to the sensors S2 to S4. The sensor S1 may also include a cylinder sensor for detecting the expansion or contraction position of the boom cylinder 7. Hereinafter, the same may apply to the sensors S2 and S3. The output of the sensor S1 (measurement data representing the attitude state of the boom 4) is received into the controller 30. Thus, the controller 30 can determine the attitude state of the boom 4.

[0124] The sensor S2 is attached to the arm 5 and measures the attitude state of the arm 5. The sensor S2 outputs measurement data indicating the attitude state of the arm 5. The attitude state of the arm 5 is, for example, the attitude angle (hereinafter, “arm angle”) around the rotational axis of

the base end corresponding to the connecting part of the arm 5 with the boom 4. The output of the sensor S2 (measurement data indicating the attitude state of the arm 5) is received into the controller 30. Thus, the controller 30 can determine the attitude state of the arm 5.

[0125] The sensor S3 is attached to the bucket 6 and measures the attitude state of the bucket 6. The sensor S3 outputs measurement data indicating the attitude state of the bucket 6. The attitude state of the bucket 6 is, for example, the attitude angle (hereinafter, “arm angle”) around the rotation axis of the base end corresponding to the connecting part with the arm 5 of the bucket 6. The output of the sensor S3 (measurement data indicating the attitude state of the bucket 6) is received into the controller 30. Thus, the controller 30 can determine the attitude state of the bucket 6.

[0126] The sensor S4 measures the attitude state of the body (e.g., upper swivel body 3) of the shovel 100. The sensor S4 outputs measurement data indicating the attitude state of the body of the shovel 100. The attitude state of the body of the shovel 100 is, for example, the inclination state of the body with respect to a predetermined reference plane (e.g., horizontal plane). For example, the sensor S4 is attached to the upper swivel body 3, and measures the inclination angle of the shovel 100 about two axes in the longitudinal direction and the lateral direction (hereinafter, “longitudinal inclination angle” and “lateral inclination angle”). The output of the sensor S4 (measurement data indicating the attitude state of the body of the shovel 100) is received into the controller 30. Thus, the controller 30 can determine the attitude state (inclination state) of the body (upper swivel body 3).

[0127] The sensor S5 is attached to the upper swivel body 3 and measures the swivel state of the upper swivel body 3. The sensor S5 outputs measurement data representing the swivel state of the upper swivel body 3. The sensor S5 measures, for example, a swivel angular velocity and a swivel angle of the upper swivel body 3. The sensor S5 includes, for example, a gyro sensor, a resolver, a rotary encoder, etc. The output of the sensor S5 (measurement data representing the swivel state of the upper swivel body 3) is received into the controller 30. Thus, the controller 30 can determine the swivel state such as the swivel angle of the upper swivel body 3.

[0128] The controller 30 can determine (estimate) the position of the tip (bucket 6) of the attachment AT based on the outputs of the sensors S1 to S5.

[0129] When the sensor S4 includes a gyro sensor capable of detecting three-axis angular velocity, a six-axis sensor, an IMU, or the like, the swivel state (e.g., swivel angular velocity) of the upper swivel body 3 may be detected based on the detection signal of the sensor S4. In this case, the sensor S5 may be omitted.

[0130] The sensor S6 measures the position of the shovel 100. The sensor S6 may measure the position in world (global) coordinates or in local coordinates at the work site. In the former case, the sensor S6 may be, for example, a GNSS (Global Navigation Satellite System) sensor. In the latter case, the sensor S6 is a transceiver capable of communicating with a device that serves as a reference for the position at the work site and outputting a signal corresponding to the position relative to the reference. The output of the sensor S6 is received into the controller 30.

[0131] The sensor S7 measures pressure (cylinder pressure) in an oil chamber of the boom cylinder 7. The sensor S7 includes, for example, a sensor for measuring the cylinder pressure (rod pressure) of the oil chamber on the rod side of the boom cylinder 7 and a sensor for measuring the cylinder pressure (bottom pressure) of the oil chamber on the bottom. The output of the sensor S7 (measurement data of the cylinder pressure of the boom cylinder 7) is received into the controller 30.

[0132] The sensor S8 measures the pressure (cylinder pressure) of the oil chamber of the arm cylinder 8. The sensor S8 includes, for example, a sensor for measuring the cylinder pressure (rod pressure) of the oil chamber on the rod side of the arm cylinder 8 and a sensor for measuring the cylinder pressure (bottom pressure) of the oil chamber on the bottom side of the arm cylinder 8. The output of the sensor S8 (measurement data of the cylinder pressure of the arm cylinder 8) is received into the controller 30.

[0133] The sensor S9 measures the pressure (cylinder pressure) of the oil chamber of the bucket cylinder 9. The sensor S9 includes, for example, a sensor for measuring the cylinder pressure (rod pressure) of the oil chamber on the rod side of the bucket cylinder 9 and a sensor for measuring the cylinder pressure (bottom pressure) of the oil chamber on the bottom side of the bucket cylinder 9. The output of the sensor S9 (measurement data of the cylinder pressure of the bucket cylinder 9) is received into the controller 30.

[0134] The controller 30 can determine the load state acting on the attachment AT based on the outputs of the sensors S7 to S9. The load acting on the attachment AT includes, for example, the reaction force acting on the bucket 6 from the target material (earth) and the weight of the earth contained in the bucket 6.

<Hardware Configuration of Information Processing Device>

[0135] FIG. 5 is a drawing illustrating an example of a hardware configuration of the information processing device 200.

[0136] The functions of the information processing device 200 are achieved by optional hardware or combinations of optional hardware and software. For example, as shown in FIG. 5, the information processing device 200 includes an external interface 201, an auxiliary storage device 202, a memory device 203, a CPU 204, a high speed arithmetic unit 205, a communication interface 206, an input device 207, a display device 208, and a sound output device 209. These devices are connected by bus BS2.

[0137] The external interface 201 functions as an interface for reading data from a storage medium 201A and writing data to the storage medium 201A. The storage medium 201A includes, for example, a flexible disk, CD (Compact Disc), DVD (Digital Versatile Disc), BD (Blu-ray (registered trademark) Disc), SD memory card, USB memory, and the like. Thus, the information processing device 200 can read various data to be used in processing through the storage medium 201A, store them in the auxiliary storage device 202, and install programs to achieve various functions.

[0138] The information processing device 200 may acquire various data and programs to be used in processing from an external device through the communication interface 206.

[0139] The auxiliary storage device 202 stores various installed programs, and also stores files, data, and the like necessary for various processing. The auxiliary storage device 202 includes, for example, hard disc drives (HDDs), solid state discs (SSDs), flash memories, and the like.

[0140] The memory device 203 reads and stores a program from the auxiliary storage device 202 when a program start instruction is given. The memory device 203 includes, for example, DRAM (Dynamic Random Access Memory) and SRAM.

[0141] The CPU 204 performs various programs loaded from the auxiliary storage device 202 to the memory device 203, and implements various functions related to the information processing device 200 according to the programs.

[0142] The high speed arithmetic unit 205 performs arithmetic processing at a relatively high speed in conjunction with the CPU 204. The high speed arithmetic unit 205 includes, for example, GPU (Graphics Processing Unit), ASIC (Application Specific Integrated Circuit), and FPGA (Field-Programmable Gate Array).

[0143] Note that, the high speed arithmetic unit 205 may be omitted depending on the required arithmetic processing speed.

[0144] The communication interface 206 is used as an interface to connect to enable to communicate with an external device. Thus, the information processing device 200 can communicate with an external device such as a shovel 100, for example, through the communication interface 206. The communication interface 206 may have a plurality of types of communication interfaces depending on a communication system or the like with the connected device.

[0145] The input device 207 receives various inputs from the user. The input device 207 includes an operating device for remote control of the shovel 100.

[0146] The input device 207 includes, for example, an input device (hereinafter, "operation input device") which accepts mechanical operation input from the user. The operating device for remote control may be an operation input device. The operation input device includes, for example, a button, a toggle, a lever, a keyboard, a mouse, a touch panel mounted on the display device 208, a touch pad provided separately from the display device 208, and the like.

[0147] The input device 207 may also include a voice input device capable of receiving voice input from a user. The voice input device may include, for example, a microphone capable of collecting user voice.

[0148] The input device 207 may also include a gesture input device capable of receiving gesture input from a user. The gesture input device may include, for example, a camera capable of capturing an image of a user's gesture.

[0149] The input device 207 may also include a biometric input device capable of receiving biometric input from a user. The biometric input device may include, for example, a camera capable of acquiring image data containing information about a user's fingerprint or iris.

[0150] The display device 208 displays an information screen and an operation screen to the user of the information processing device 200. The display device 208 is, for example, a liquid crystal display or an organic EL (Electroluminescence) display.

[0151] The sound output device 209 transmits various types of information by sound to the user of the information

processing device **200**. The sound output device **209** is, for example, a buzzer, alarm, speaker, etc.

[Functional Configuration of Operation Support System]

[0152] Next, a functional configuration of the operation support system SYS will be described with reference to FIGS. **6** and **7** in addition to FIGS. **1** to **5**.

First Example

[0153] FIG. **6** is a functional block diagram illustrating a first example of the functional configuration of the operation support system SYS.

[0154] Hereinafter, a term “path of the work part of the shovel **100**” includes both a route where the work part of the shovel **100** has already moved (i.e., path) and a route where it may move in the future. The work part corresponds to the tip of the attachment AT used for applying a change with respect to the target material. Specifically, the work part is the bucket **6**.

[0155] The shovel **100** includes a support device **150**. In this example, the support device **150** supports the shovel **100** operated by the autonomous operation function in performing the work.

[0156] As shown in FIG. **6**, in this example, the support device **150** includes a controller **30**, a hydraulic controlling valve **31**, a sensor **40**, an output device **50**, an input device **52**, and sensors S1 to S9. Further, when instructions concerning autonomous operation are input from outside the shovel **100**, the support device **150** may include a communication device **60** instead of or in addition to the input device **52**.

[0157] The controller **30** includes, as functional parts, an operation log providing part **301** and a work support part **302**.

[0158] When there are multiple shovels **100** included in the operation support system SYS, there may be a shovel **100** in which the controller **30** includes only the operation log providing part **301** and the work support part **302**, and a shovel **100** in which only the latter is included. In this case, the former shovel **100** has only the function of acquiring an operation log of the shovel **100** used for the work support function of the latter shovel **100** and providing it to the information processing device **200**. The same may be applied to the case of the second example described later.

[0159] The information processing device **200** includes, as functional parts, a log acquisition part **2001**, a simulator part **2002**, a log storage part **2003**, a training data generation part **2004**, a machine learning part **2005**, a trained model storage part **2006**, and a distribution part **2007**.

[0160] The operation log providing part **301** is a functional part for acquiring the operation log of a predetermined operation of the shovel **100** and providing it to the information processing device **200**.

[0161] For each type of work, a plurality of (types of) predetermined operations are previously defined. For example, in the case of excavation work, the plurality of predetermined operations include an excavation operation, a boom raising and swivel operation, a boom lowering and swivel operation, an earth removing operation, a broom operation, and the like. In the case of ground leveling work, the plurality of predetermined operations include an excavation operation, an earth removing operation, a sweeping operation, a horizontal pulling operation, a rolling operation,

a broom operation, and the like. In the case of slope work, the plurality of predetermined operations may include an excavation operation, an earth removing operation, a slope pulling operation, a rolling operation, and the like. The slope pulling operation corresponds to the horizontal pulling operation of the ground leveling work, and involves moving the attachment AT to draw a cutting edge (edge of teeth) of the bucket **6** toward the body (upper swivel body **3**) along the slope corresponding to the target construction surface. The sweeping operation involves, for example, operating the attachment AT and pushing the bucket **6** forward along the ground to sweep the earth forward at the back of the bucket **6**.

[0162] In the sweeping operation, for example, the attachment AT performs a lowering operation of the boom **4** and an opening operation of the arm **5**. The horizontal pulling operation involves, for example, operating the attachment AT and moving the edge of teeth of the bucket **6** along the ground drawing toward the front approximately horizontally to level unevenness of the ground (surface of terrain). In the horizontal pulling operation, for example, the attachment AT performs a raising operation of the boom **4** and a closing operation of the arm **5**. The rolling operation involves, for example, operating the attachment AT and pressing the ground against the back of the bucket **6**. Moreover, the rolling operation may involve pressing the ground by striking the back of the bucket **6** against the ground while moving the bucket **6** up and down. Further, the rolling operation may involve pushing the ground against the back of the bucket **6** forward along the ground so that the earth is swept out to a predetermined position forward by the back of the bucket **6**, and then pressing the ground against the ground with the back of the bucket **6**. In the rolling operation, for example, the attachment AT performs a lowering operation of the boom **4** when pressing the ground. The broom operation involves, for example, operating the upper swivel body **3** to swivel the bucket **6** left and right while keeping it along the ground. The broom operation may involve operating the attachment AT and the upper swivel body **3** to swivel the bucket **6** left and right while keeping it along the ground and push forward the bucket **6**. In the broom operation, for example, the upper swivel body **3** alternately repeats left and right swivel operations. In the broom operation, for example, in addition to the alternating left and right swivel operation of the upper swivel body **3**, the attachment AT may perform the lowering operation of the boom **4** and the opening operation of the arm **5**.

[0163] The operation log of the shovel **100** is time-series data representing the operation state of the shovel **100**. For example, the operation log of the shovel **100** includes time-series data representing the operation contents of the operator. The time-series data representing the operation contents of the operator includes, for example, time-series output data of the operating pressure sensor **29** corresponding to the hydraulic pilot type operating device **26** and time-series output data (operation signal data) of the operating device **26** corresponding to the electric type operating device **26**. The operation log of the shovel **100** may be time-series output data of the sensors S1 to S5 and time-series data representing the attitude state of the shovel **100** obtained from the output data of the sensors S1 to S5.

[0164] In addition, the operation log providing part **301** may acquire the operation log of the shovel **100** operated by an operator (hereinafter, “skilled person” for convenience)

having a long operating history of the shovel 100 and provide the operation log to the information processing device 200. Thus, the trained model LM3 capable of reproducing the operation of the shovel 100 operated by an expert can be generated by machine learning based on the operation log of the shovel 100, as described later.

[0165] The operation log providing part 301 includes an operation log recording part 301A, an operation log storage part 301B, and an operation log transmission part 301C.

[0166] An operation log recording part 301A acquires an operation log during a predetermined operation of the shovel 100 and records it in an operation log storage part 301B. For example, each time a predetermined operation of the shovel 100 is performed, the operation log recording part 301A records an operation log during the operation in an operation log storage part 301B.

[0167] The operation log storage part 301B stores the operation log of the shovel 100. For example, for each predetermined operation performed by the shovel 100, the operation log and data of the time (date and time) when the predetermined operation was performed are associated and stored in the operation log storage part 301B. The data of the time when the predetermined operation was performed includes data of both the start and the end of the predetermined operation of the shovel 100. When a plurality of predetermined operations are defined, for each predetermined operation performed by the shovel 100, the operation log, data of the time when the predetermined operation was performed, and data of identification information of the performed predetermined operation are associated and stored in the operation log storage part 301B. Hereinafter, data linked to the operation log of the shovel 100 may be conveniently referred to as “accompanying data”. For example, for each predetermined operation performed by the shovel 100, record data representing a correspondence between the operation log and the accompanying data accumulated in the operation log storage part 301B, so that a database of the operation log at the time of performance of the predetermined operation of the shovel 100 is constructed.

[0168] The operation log of the operation log storage part 301B which has already been transmitted to the information processing device 200 by the operation log transmission part 301C described later may be deleted afterwards.

[0169] The operation log transmission part 301C transmits the operation log stored in the operation log storage part 301B when the shovel 100 performs a predetermined operation and the accompanying data associated with the operation log to the information processing device 200 through the communication device 60. The operation log transmission part 301C may also transmit to the information processing device 200 record data showing the correspondence between the operation log of the shovel 100 and the accompanying data for each predetermined operation performed by the shovel 100.

[0170] For example, the operation log transmission part 301C transmits a non-transmitted operation log of the shovel 100 and the accompanying data stored in the operation log storage part 301B to the information processing device 200 in response to a transmission request of the operation log of the shovel 100 received from the information processing device 200. The operation log transmission part 301C may also automatically transmit the non-transmitted operation log of the shovel 100 and the accompanying data stored in

the operation log storage part 301B to the information processing device 200 at a predetermined timing. The predetermined timing is, for example, when the shovel 100 stops operation (key switch is turned off) or starts operation (key switch is turned on).

[0171] The log acquisition part 2001 acquires a log when the shovel 100 performs a predetermined operation.

[0172] The log when the shovel 100 performs the predetermined operation includes the operation log when the shovel 100 performs the predetermined operation and the state log of the target material. The state log of the target material includes time-series data showing the state of the target material before, during, and after the performance of the predetermined operation of the shovel 100. The state of the target material includes a shape of the earth (terrain shape) and a characteristic of the earth of the target material, and the like. The characteristics of the earth may include, for example, hardness of the earth, moisture content of the earth, size of the earth grain (grain size), angle of repose of the earth, and the like. The operation log when the shovel 100 performs the predetermined operation is uploaded from the shovel 100. The state log of the target material when the shovel 100 performs the predetermined operation is acquired based on the measurement data uploaded from the sensor group 300 and the accompanying data (data at the time when the predetermined operation is performed) uploaded from the shovel 100.

[0173] The state log of the target material may be acquired based on the measurement data of the sensor 40 of the shovel 100. In this case, the measurement data acquired by the sensor 40 at the time of the predetermined operation of the shovel 100 is uploaded from the shovel 100 to the information processing device 200. In this case, the sensor group 300 may be omitted.

[0174] The simulator part 2002 performs computer simulation of a predetermined operation of the shovel 100 using a virtual model of the shovel 100 and the target material (earth).

[0175] For example, by using a Distinct Element Method (DEM), the earth of the target material is modeled as a collection of minute particles. Thus, the simulator part 2002 causes the virtual model of the shovel 100 to perform a predetermined operation such as an excavation operation, and can virtually reproduce the whole behavior of the earth of the target material and the reaction forces from the earth in entirety by analyzing the individual operation of the minute particles.

[0176] The simulator part 2002 acquires data on the path of the work part of the shovel 100 and data on the state of the target material (earth) before, during, and after the performance of the predetermined operation as a log when the shovel 100 performs the predetermined operation by computer simulation. The former data corresponds to the operation log when the shovel 100 performs the predetermined operation by computer simulation, and the latter data corresponds to the state log of the target material when the shovel 100 performs the predetermined operation by computer simulation.

[0177] The simulator part 2002 performs computer simulation of a large number of patterns concerning a predetermined operation of the shovel 100 using various conditions of target materials (earth) and paths of various work parts of the shovel 100. Thus, the simulator part 2002 can store logs

when the shovel **100** performs a predetermined operation by computer simulation under different conditions in the log storage part **2003**.

[0178] Logs when the shovel **100** performs a predetermined operation acquired by the log acquisition part **2001** and the simulator part **2002** are stored in the log storage part **2003** in an accumulated form. For example, the log storage part **2003** stores an operation log for each predetermined operation actually performed by the shovel **100** or by computer simulation, a state log of the target material, and accompanying data in a form linked to each other. In the log storage part **2003**, the logs acquired by the log acquisition part **2001** and the logs acquired by the simulator part **2002** may be stored in a distinguishable manner or may be stored in a mixed manner in an indistinguishable manner.

[0179] The training data generation part **2004** generates training data for machine learning based on the logs when the shovel **100** performs a predetermined operation stored in the log storage part **2003**, and outputs a training data set which is an aggregate of a large number of training data. The training data generation part **2004** may automatically generate the training data by batch processing, or may generate the training data in response to an input from the user of the information processing device **200**. The training data generation part **2004** includes training data generation parts **2004A** to **2004C**.

[0180] The training data generation part **2004A** generates a training data set for generating the trained model LM1. The trained model LM1 infers a future state of the target material of the shovel **100** at a predetermined time in the future by using a current state of the target material of the shovel **100** and the path of the work part of the shovel **100** up to a predetermined time in the future as inputs. The training data is a combination of the state of the target material of the shovel **100** at a first time and a path of the work part of the shovel **100** from the first time to a second time after the first time as input data and the state of the target material at the second time as ground truth.

[0181] Note that the training data set for generating the trained model LM1 may be generated only from the former log among the log acquired by the log acquisition part **2001** and the log output from the simulator part **2002**. In this case, the simulator part **2002** may be omitted. Similarly, the training data set for generating the trained model LM1 may be generated only from the latter log among the log acquired by the log acquisition part **2001** and the log output from the simulator part **2002**. In this case, the operation log providing part **301** of the sensor group **300** and the shovel **100** may be omitted. In addition, the training data set for generating the trained model LM1 may include a base training data set and a final adjustment (fine-tuning) training data set. In this case, since a large number of data is required, the base training data set may be generated based on the log output from the simulator part **2002**, and the final adjustment training data set may be generated based on the log acquired by the log acquisition part **2001**. Hereinafter, the same may be applied to the trained models LM2 and LM3.

[0182] The training data generation part **2004B** generates training data for generating the trained model LM2. When a plurality of tasks are defined, the trained model LM2 is generated for each of the plurality of tasks. The trained model LM2 uses data of the state of the target material around the shovel **100** as an input, and infers one predetermined operation suitable for the state of the target material

corresponding to the input data from among the plurality of predetermined operations used in the work.

[0183] The training data is, for example, a combination of the state of the target material before the performance of the predetermined operation of the shovel **100** as input data and the type of the predetermined operation performed by the shovel **100** thereafter as ground truth. The training data may further include the target shape (e.g., target construction surface) of the target material as input data. The training data generation part **2004B** may generate a training data set based on the log obtained by the log acquisition part **2001** when the shovel **100** performs the predetermined operation by a skilled person's operation. Thus, the trained model LM2 can reproduce the way in which the skilled person selects a prescribed operation of the shovel **100**.

[0184] A training data generation part **2004C** generates training data for generating the trained model LM3. The trained model LM3 is used for inferring a target path of a work part in a prescribed operation of the shovel **100** with data of the state of a target material around the shovel **100** as input. The trained model LM3 is generated for each (type of) prescribed operation of the shovel **100**.

[0185] The trained model LM3, for example, infers an operation parameter that defines a target path of a work part in a prescribed operation of the shovel **100** based on the state of the target material before the performance of the prescribed operation of the shovel **100**. In this case, the training data is a combination of the state of the target material before the performance of the prescribed operation of the shovel **100** as input data and the operation parameter corresponding to the path of the work part when the shovel **100** performs the prescribed operation as ground truth. The trained model LM3 may also infer a target path of a work part in a prescribed operation of the shovel **100** based on the state of the target material before the performance of the prescribed operation of the shovel **100**. In this case, the training data is a combination of the state of the target material before the performance of the prescribed operation of the shovel **100** as input data and the path of the work part when the shovel **100** performs the prescribed operation as ground truth. The training data may further include the target shape (e.g., target construction surface) of the target material as input data. In addition, the training data generation part **2004C** may generate a training data set based on a log obtained by the log acquisition part **2001** when the shovel **100** performs a predetermined operation by an expert. Thus, the trained model LM3 can reproduce the operation of the shovel **100** by the expert.

[0186] The machine learning part **2005** causes the base learning model to perform machine learning based on the training data set generated by the training data generation part **2004**, and generates the trained models LM1 to LM3. The trained model (the base learning model) includes a neural network such as a DNN (Deep Neural Network), for example.

[0187] The machine learning part **2005** includes machine learning parts **2005A** to **2005C**.

[0188] The machine learning part **2005A** causes the base learning model M1 to perform machine learning based on the training data set output from the training data generation part **2004A**. Thus, the machine learning part **2005A** can generate a trained model LM1 capable of outputting (inferring) the state of the target material of the shovel **100** at a predetermined time in the future by taking data such as the

current state of the target material of the shovel **100** and the target path of the work part of the shovel **100** up to a predetermined time in the future as inputs. Further, the machine learning part **2005A** may correct (additionally train) the trained model **LM1** so that an error between the inference result of the trained model **LM1** and the measurement result of an actual sensor **40** is reduced. In this case, the inference result of the trained model **LM1** and the data of the measurement result of the actual sensor **40** are uploaded from the shovel **100** to the information processing device **200**.

[0189] The machine learning part **2005B** causes the base learning model **M2** to perform machine learning based on the training data set output from the training data generation part **2004B**. Thus, the machine learning part **2005B** can generate the trained model **LM2** capable of outputting (inferring) one predetermined operation from among a plurality of predetermined operations corresponding to the target operation using data of the state of the target operation around the shovel **100** as an input. Further, the machine learning part **2005B** may generate the trained model **LM2** by performing reinforcement learning using the simulator part **2002**.

[0190] The machine learning part **2005C** causes the base learning model **M3** to perform machine learning based on the training data set output from the training data generation part **2004C**. Thus, the machine learning part **2005C** can generate the trained model **LM3** capable of outputting (inferring) the target path of the target operation in the predetermined operation of the shovel **100** using data of the state of the target operation around the shovel **100** as an input.

[0191] Trained models **LM1** and **LM2** output by the machine learning part **2005** are stored in the trained model storage part **2006**. When the trained model **LM1** is re-trained or additionally trained by the machine learning part **2005A**, the trained model **LM1** in the trained model storage part **2006** is updated. The same applies when the trained models **LM2** and **LM3** are re-trained or additionally trained by the machine learning parts **2005B** and **2005C**.

[0192] The distribution part **2007** distributes the data of the trained models **LM1** to **LM3** to the shovel **100**. For example, when the trained model **LM1** is generated or updated by the machine learning part **2005A**, the distribution part **2007** distributes the most recently generated or updated trained model **LM1** to the shovel **100**. The distribution part **2007** may also distribute the latest trained model **LM1** of the trained model storage part **2006** to the shovel **100** in response to a signal received from the shovel **100** requesting the distribution of the trained model **LM1**. The same may be applied to the trained models **LM2** and **LM3**.

[0193] The work support part **302** is a functional part for supporting the work of the shovel **100** operating by the autonomous operation function.

[0194] The work support part **302** includes a trained model storage part **302A**, a target material state prediction part **302B**, an operation planning part **302C**, a target path generation part **302D**, and an operation control part **302E**.

[0195] The trained model storage part **302A** stores trained models **LM1** and **LM2** distributed from the information processing device **200** received and through the communication device **60**.

[0196] The target material state prediction part **302B** predicts the state of the target material of the shovel **100** at a

predetermined time in the future based on the current state of the target material around the shovel **100** and the target path of the work part of the shovel **100** up to a predetermined time. Specifically, the target material state prediction part **302B** predicts the state of the target material of the shovel **100** at a future time using the trained model **LM1**.

[0197] The current state of the target material around the shovel **100** is acquired, for example, based on the output of the sensor **40**. The current state of the target material around the shovel **100** may be acquired based on the output of the sensors **S7** to **S9** instead of or in addition to the sensor **40**. This is because the controller **30** can estimate the reaction force from the ground acting on the bucket **6** from the output of the sensors **S7** to **S9**, and can estimate the state (shape and characteristics) of the earth of the target material from the estimated result of the reaction force. The current state of the target material around the shovel **100** may be a prediction result of the state of the target material output by the target material state prediction part **302B** itself at a processing timing earlier than the current time. In this case, for example, the target material state prediction part **302B** predicts the state of the target material based on the initial state of the target material at the beginning of the work, and subsequently uses the prediction result as the current state of the target material. The initial state of the target material at the beginning of the work may be provided from the outside of the shovel **100**, or may be previously defined as a fixed state, such as a plane at the same height as the ground on which the lower traveling body **1** of the shovel **100** is grounded.

[0198] The predetermined time point is, for example, the timing (time t_b) of the start of the predetermined operation that can be modified immediately. A modification of the predetermined operation means, for example, the modification of the type of the predetermined operation that is scheduled to be performed to another type, such as the modification of a future predetermined in operation an undetermined or tentatively determined state or an excavation operation that is scheduled to be performed to a slope pulling operation. The predetermined time point may be the timing (time t_s) at which the path of the bucket **6** and the type of the predetermined operation can be modified, considering the delay time τ_s in the process. The delay time τ_s includes, for example, the calculation time for the controller **30** to generate the target path of the bucket **6** or to determine the predetermined operation of the shovel **100**, and the interface time to transfer the calculation result to the control part.

[0199] The time t_s is calculated by the following equation (1) using the current time t_1 and the delay time τ_s .

[Math 1]

$$t_s = t_1 + \tau_s \quad (1)$$

[0200] The delay time τ_s may be a fixed value or a variable value. In the former case, the fixed value is previously defined as the maximum value of the delay time that can be assumed depending on, for example, the processing state of the controller **30**. In the latter case, the delay time τ_s is varied according to a predetermined rule depending on, for example, the processing state of the controller **30** such as the CPU load state.

[0201] The time t_b corresponds to a start time of the next predetermined operation of the predetermined operation being performed at the time t_s .

[0202] For example, as shown in FIG. 7, when the time t_s is before the start time of the next operation B of the current operation A of the shovel 100, the time t_b corresponds to the start time of the next operation B of the current operation A.

[0203] Conversely, as shown in FIG. 8, when the time t_s is after the start time of the next operation B of the current operation A of the shovel 100, the controller 30 cannot modify the next operation B to another type of predetermined operation due to the delay time τ_s . Therefore, in this case, the time t_b corresponds to the start time of the next operation C of the operation B being performed at the time t_s .

[0204] The operation planning part 302C plans (determines) the predetermined operation (type) to be performed by the shovel 100 from the time t_b based on the prediction result (prediction result of the state of the target material at the time t_b) by the target material state prediction part 302B.

[0205] Thus, the controller 30 can determine the predetermined operation type to be performed by the shovel 100 in accordance with the predicted state of the target material in the future (time t_b). Therefore, for example, it is not necessary to stop the operation of the shovel 100 to some extent, as in the case of determining the subsequent predetermined operation based on the actual state of the target material at the time of completion of the predetermined operation immediately before the time t_b . Therefore, the controller 30 can improve the work efficiency of the shovel 100.

[0206] The operation planning part 302C may adjust the processing timing so as to plan the next predetermined operation after the predetermined operation currently being performed by the shovel 100. For example, when the remaining time until the end time of the predetermined operation currently being performed by the shovel 100 is not less than the delay time τ_s , the operation planning part 302C plans the next predetermined operation of the shovel 100.

[0207] For example, the operation planning part 302C determines the predetermined operation (type) to be started at the time t_b based on the prediction result of the state of the target material at the time t_b by using a rule-based method.

[0208] For example, a plurality of predetermined operations that can be performed are defined for each of the target materials, and transition conditions for each of the previously defined operations that can transition are specified in advance for each of the predetermined operations. The predetermined operations that can transition can include the same predetermined operation. This is because the same predetermined operation may be repeated. Then, the operation planning part 302C determines the predetermined operation (type of operation) to be started at time t_b based on the success or failure of a plurality of transition conditions starting from the predetermined to be performed operation immediately before time t_b .

[0209] For example, as shown in FIG. 9, in the slope work of this example, the excavation operation ST1-1, the earth removing operation ST1-2, and the slope pulling operation ST1-3 are defined as the plurality of predetermined operations that can be performed.

[0210] The predetermined operations that can be transitioned from a standby state ST1-0 corresponding to before the work start and after the work completion are the excavation

operation ST1-1 and the slope pulling operation ST1-3, and the transition conditions SC1-01 and SC1-03 are defined respectively. The transition conditions SC1-01 and SC1-03 are mutually opposing conditions. For example, if a difference between the shape of the target material and the target shape at the beginning of the work is not more than the predetermined standard, the transition condition SC1-03 is established, and if it exceeds the predetermined standard, the transition condition SC1-01 is established. The predetermined operation to be performed at the beginning of the work is determined to be the predetermined operation corresponding to one of the transition conditions SC1-01 and SC1-03.

[0211] The predetermined operation that can transition from the excavation operation ST1-1 is the earth removing operation ST1-2, and the transition condition SC1-12 is defined. That is, when the predetermined operation performed immediately before time t_b is the excavation operation ST1-1, the transition condition SC1-12 is always established, and the predetermined operation performed from time t_b is unambiguously determined to be the earth removing operation ST1-2.

[0212] The predetermined operations that can transition from the earth removing operation ST1-2 are the excavation operation ST1-1 and the slope pulling operation ST1-3, and the transition conditions SC1-21 and SC1-23 are defined respectively. The transition conditions SC1-21 and SC1-23 are mutually opposing conditions. For example, when the difference between the predicted result of the shape of the target material and the target shape at time t_b is not more than the predetermined standard, the transition condition SC1-23 is established, and when it exceeds the predetermined standard, the transition condition SC1-21 is established. When the predetermined operation to be performed immediately before time t_b is the earth removing operation ST1-2, the predetermined operation to be performed from time t_b is determined to be the predetermined operation corresponding to one of the transition conditions SC1-21 and SC1-23.

[0213] The predetermined operations that can transition from the slope pulling operation ST1-3 are the excavation operation ST1-1 and the slope pulling operation ST1-3, and the transition conditions SC1-31 and SC1-33 are defined respectively. The transition conditions SC1-31 and SC1-33 are mutually opposite. If the predetermined operation performed immediately before time t_b is the slope pulling operation ST1-3, the predetermined operation performed from time t_b is determined to be the predetermined operation corresponding to one of the transition conditions SC1-31 and SC1-33.

[0214] If the work completion condition indicating that the work has been completed is satisfied at time t_b , the controller 30 shifts to the standby state without performing the predetermined operation determined in advance by the operation planning part 302C (see a dashed arrow in the figure). For example, the work completion condition is that the difference between the shape of the target material and the target shape just before time t_b is so small that it can be considered zero.

[0215] Further, the operation planning part 302C may use the trained model LM2 to determine the predetermined operation (type of operation) to be started by the shovel 100 at time t_b based on the prediction result of the state of the target material at time t_b .

[0216] For example, as shown in FIG. 10, in the case of the ground leveling operation, there are many predetermined operations that can be performed and the combination of transition destinations for each predetermined operation is complicated. As a result, there is a possibility that the transition condition starting from a certain predetermined operation cannot be properly set. However, by using the trained model LM2, even in the case of the ground leveling operation, where there are many predetermined operations that can be performed and the combination of transition destinations for each predetermined operation is complicated, the operation planning part 302C can more appropriately determine the predetermined operation to be started by the shovel 100 at time t_b .

[0217] When the operation planning part 302C can determine the predetermined operation to be performed by the shovel 100 from time t_b only by the rule-based method, the training data generation part 2004B and the machine learning part 2005B are omitted.

[0218] The target path generation part 302D generates the target path of the work part in the predetermined operation of the shovel 100 based on the state of the target material around the shovel 100. The predetermined operation at this time is the type of predetermined operation determined by the operation planning part 302C.

[0219] For example, the target path generation part 302D generates the target path of the work part after time t_s using the trained model LM3 based on the prediction result (predicted state of the target material at time t_s , t_b) of the target material state prediction part 302B. Thus, for example, the controller 30 can modify the target path of the work part in the predetermined operation of the shovel 100 in accordance with the state (prediction result) of the target material during performance of the predetermined operation. Therefore, the shovel 100 can progress the work of the shovel 100 more appropriately and efficiently according to the change of the state of the target material.

[0220] Specifically, the target path generation part 302D may generate the target path of the work part in the predetermined operation performed at time t_s and the target path of the work part in the predetermined operation started from time t_b .

[0221] The target path generation part 302D may generate the target path of the work part of the shovel 100 in accordance with the state (prediction result) of the target material around the shovel 100 by applying any known technique instead of the trained model LM3. In this case, the training data generation part 2004C and the machine learning part 2005C may be omitted. For example, the target path generation part 302D may generate data of the target path of the work part of the shovel 100 by MPC (Model Predictive Control) based on the prediction result (predicted state of the target material at time t_s , t_b) of the target material state prediction part 302B. In addition, the target path generation part 302D may generate data of the target path of the work part of the shovel 100 by optimizing the previously defined reference path of the work part of the shovel 100 based on data concerning the characteristics of earth given in advance.

[0222] The operation control part 302E causes the shovel 100 to perform a predetermined operation so that the predetermined part of the shovel 100 moves along the target path generated by the target path generation part 302D. Specifically, the operation control part 302E causes the shovel 100 to perform a predetermined operation so that the

work part of the shovel 100 moves along the target path by controlling the hydraulic controlling valve 31 while determining the position of the work part from the outputs of the sensors S1 to S5. Thus, the shovel 100 can autonomously advance the work while performing the predetermined operation according to the shape of the target material.

[0223] Thus, in this example, the controller 30 determines a future predetermined operation of the shovel 100 from among a plurality of predetermined operations corresponding to the target work according to the performance status of the predetermined operation of the working machine. Specifically, the controller 30 may predict the future state of the target material according to the performance status of the operation of the working machine. Then, the controller 30 may determine a future predetermined operation of the shovel 100 from among a plurality of predetermined operations corresponding to the target work based on the prediction result of the future state of the target material. Thus, the controller 30 can determine in advance the operation of the shovel 100 to be performed in accordance with the future state of the target material. Therefore, idle time when the predetermined operation of the shovel 100 is completed and the next predetermined operation is performed can be reduced, and the work efficiency of the shovel 100 can be improved.

[0224] Moreover, in this example, after determining the future predetermined operation of the shovel 100, the controller 30 generates a target path of the work part corresponding to the determined predetermined operation of the shovel 100. Thus, the controller 30 can hierarchically determine the future predetermined operation of the shovel 100 and generate a target path of the work part in the future predetermined operation of the shovel 100. Therefore, for example, when the operation plan of the shovel 100 and the path plan of the work part of the shovel 100 are performed in parallel, it is possible to control against an occurrence of a situation in which the operation plan and the path plan cannot be performed in a realistic time as a result of enormous conditions and parameters.

[0225] Some or all of the functions of the target material state prediction part 302B, the operation planning part 302C, the target path generation part 302D, and the operation control part 302E may be transferred to the information processing device 200. Thus, the processing load of the shovel 100 can be reduced for the processing related to the generation of the target path of the work part of the shovel 100 and the processing related to the control of the operation of the shovel 100.

Second Example

[0226] FIG. 11 is a functional block diagram illustrating a second example of the functional configuration of the operation support system SYS.

[0227] Hereinafter, the same reference numerals are assigned to the same or corresponding configurations as in the first example described above, and the description will be made mainly on the different parts from the first example described above.

[0228] The shovel 100 includes a support device 150 as in the first example described above. In this example, the support device 150 supports a user who operates the semi-automatic shovel 100 and performs work.

[0229] As shown in FIG. 11, in this example, as in the first example described above, the support device 150 includes a

controller 30, a hydraulic controlling valve 31, a sensor 40, an output device 50, and sensors S1 to S9. When the shovel 100 is remotely operated, the support device 150 may include a communication device 60 instead of or in addition to the input device 52.

[0230] As in the first example described above, the controller 30 includes, as functional parts, an operation log providing part 301 and a work support part 302.

[0231] As in the first example described above, the operation log providing part 301 includes, as functional parts, an operation log recording part 301A, an operation log storage part 301B, and an operation log transmission part 301C.

[0232] As in the first example described above, the information processing device 200 includes, as functional parts, a log acquisition part 2001, a simulator part 2002, a log storage part 2003, a training data generation part 2004, a machine learning part 2005, a trained model storage part 2006, and a distribution part 2007.

[0233] The work support part 302 is a functional part for supporting a user who operates the semi-automatic shovel 100 and performs work.

[0234] As in the first example described above, the work support part 302 includes a trained model storage part 302A, a target material state prediction part 302B, an operation planning part 302C, a target path generation part 302D, and an operation control part 302E. Unlike the first example described above, the work support part 302 includes an operation proposal part 302F.

[0235] The operation proposal part 302F proposes predetermined operation (type) of the shovel 100 from time t_b determined (planned) by the operation planning part 302C to the user through the output device 50 and the remote control support device 400.

[0236] In response to the proposal of the predetermined operation of the shovel 100 by the operation proposal part 302F, the user selects a predetermined operation (type) to be performed by the shovel 100 starting from time t_b from among a plurality of predetermined operations corresponding to the current work. At this time, the user selects a predetermined operation to be performed by the shovel 100 starting from time t_b using the input device 52 and the remote control support device 400, for example. The selection result by the user is input to the target path generation part 302D through the input device 52 and the communication device 60.

[0237] The target path generation part 302D generates the target path of the work part after time t_s by using the trained model LM3 based on the prediction result of the target material state prediction part 302B (the prediction result of the state of the target material at time t_s).

[0238] Specifically, the target path generation part 302D may generate the target path of the work part in the predetermined operation performed at time t_s and the target path of the work part in the predetermined operation started at time t_b . In this case, the predetermined operation started at time t_b is the predetermined operation of the shovel 100 corresponding to the selection result by the user.

[0239] Thus, in this example, the controller 30 determines the future predetermined operation of the shovel 100 from among a plurality of predetermined operations corresponding to the target operation based on the prediction result of the state of the target material in the future, and proposes it to the user through the output device 50 and the remote control support device 400. Thus, the user can determine the

type of a recommended predetermined operation of the shovel 100 based on the prediction result of the state of the target material in the future before the completion of the predetermined operation immediately before the recommended predetermined operation of the shovel 100. Therefore, idle time when the predetermined operation of the shovel 100 is completed and the next predetermined operation is performed can be reduced, and the work efficiency of the shovel 100 can be improved.

[0240] The functions of the operation planning part 302C and the operation proposal part 302F may be adopted in a manually operated shovel 100, whose operations are entirely controlled by the operator. In this case, the controller 30 may predict the path of the work part in the predetermined operation of the shovel 100 based on the history of the operation contents of the operator or the output of the sensors S1 to S9, and may predict the future state of the target material based on the predicted path. When the shovel 100 is operated remotely, some or all of the functions of the trained model storage part 302A, the target material state prediction part 302B, the operation planning part 302C, the target path generation part 302D, the operation control part 302E, and the operation proposal part 302F may be provided in the remote control support device 400. In addition, some or all of the functions of the target material state prediction part 302B, the operation planning part 302C, the target path generation part 302D, the operation control part 302E, and the operation may be transferred to the proposal part 302F information processing device 200. Thus, the processing load of the shovel 100 and the remote control support device 400 can be reduced in the processing related to the generation of the target path of the work part of the shovel 100 and the processing related to the control of the operation of the shovel 100.

[Specific Example of Processing Related to Autonomous Operation of Shovel]

[0241] Next, a specific example of processing related to the autonomous operation of the shovel 100 will be described with reference to FIGS. 12 to 17.

[0242] In this example, the description proceeds assuming the functional configuration of the operation support system SYS shown in FIG. 6.

<Process Related to Start of Autonomous Operation of Shovel>

[0243] FIG. 12 is a flowchart schematically illustrating an example of processing related to a start of autonomous operation of the shovel 100. This flowchart is performed when a predetermined input related to the start of autonomous operation is made by the user through the input device 52, the remote control support device 400, or the remote monitoring support device.

[0244] In a step S102, the controller 30 selects a predetermined operation to be performed at the beginning of autonomous operation of the shovel 100 in response to a predetermined input from the user through the input device 52, the remote control support device 400, or the remote monitoring support device. For example, the controller 30 selects one predetermined operation from among a plurality of predetermined operations defined for the target work such as excavation work, leveling work, and slope work. Further, prior to the step S102, the controller 30 may select one

operation from among a plurality of operations such as excavation work, leveling work, and slope work in response to a predetermined input from the user through the input device **52**, the remote control support device **400**, or the remote monitoring support device.

[0245] When the processing of the step **S102** is completed, the controller **30** proceeds to a step **S104**.

[0246] In the step **S104**, the controller **30** acquires data representing the state (shape and characteristic) of the earth to be worked, based on the output of the sensor **40**.

[0247] When the processing of the step **S104** is completed, the controller **30** proceeds to a step **S106**.

[0248] In the step **S106**, the controller **30** (target path generation part **302D**) generates the target path of the bucket **6** for the predetermined operation of the shovel **100** selected in the step **S102** at the beginning of the work, based on the data acquired in the step **S104**. For example, the target path generation part **302D** generates the target path of the work part of the shovel **100** in the same manner as in a step **S208** described later.

[0249] When the processing of the step **S106** is completed, the controller **30** proceeds to the step **S108**.

[0250] In the step **S108**, the controller **30** notifies the user that autonomous operation is enabled. For example, the controller **30** notifies the user inside the cabin **10** or the user near the shovel **100** through the output device **50**. The controller **30** may also notify the user using the remote control support device **400** or the remote monitoring support device by transmitting the notification signal to the remote control support device **400** or the remote monitoring support device through the communication device **60**.

[0251] When the processing in the step **S108** is completed, the controller **30** proceeds to a step **S110**.

[0252] In the step **S110**, the controller **30** starts the autonomous operation of the shovel **100** in response to an instruction from the user received through the input device **52**, the remote control support device **400**, or the remote monitoring support device.

[0253] When the processing of the step **S110** is completed, the controller **30** ends processing of the present flowchart.

[0254] Thus, the controller **30** can start the autonomous operation of the shovel **100** in response to a predetermined input from the user.

<Process Related to Bucket Path Generation (Main Flow)>

[0255] FIG. **13** is a main flowchart schematically illustrating an example of processing related to bucket path generation. FIG. **14** is a drawing illustrating an example of an observation target area **TA**.

[0256] This flowchart is repeatedly performed every predetermined control cycle after the start of the autonomous operation of the shovel **100**.

[0257] In a step **S202**, the controller **30** acquires the future times t_s and t_b as references for the operation plan of the shovel **100** and the path generation.

[0258] When the processing in the step **S202** is completed, the controller **30** proceeds to a step **S204**.

[0259] In the step **S204**, the target material state prediction part **302B** predicts the state of the earth of the target material (ground) at times t_s and t_b . Specifically, the target material state prediction part **302B** predicts the state of the earth of the target material at a modifiable start time t_s based on the state of the earth of the target material at the current time t_1 and the target path of the bucket **6** from the current time t_1

to the time t_s . Similarly, the target material state prediction part **302B** predicts the state of the earth of the target material at time t_b based on the state of the earth of the target material at the current time t_1 and the target path of the bucket **6** from the current time t_1 to the time t_b . The data of the target path of the bucket **6** used in this step is obtained in the process of a step **S208** in the previous flow chart or in the process of the step **S106** in FIG. **8**.

[0260] For example, as shown in FIG. **14**, the observation target area **TA** around the shovel **100** is divided into grid units of a predetermined number N . The observation target area **TA** is an area around the shovel **100** in which the target material state prediction part **302B** acquires data indicating the state of the earth. In this example, at time t , a shape h' of the earth and a characteristic κ' of the earth of each grid unit i ($i=1$ to N) of the observation target area **TA** are defined.

[0261] For example, the shape h' of the earth and the characteristic κ' of the earth of the work target of the shovel **100** for each grid unit i of the observation target area **TA** at the current time t_1 are expressed by the following equations (2) and (3).

[Math 2]

$$h' = [h'_1, h'_2, \dots, h'_N]^T \quad (2)$$

$$\kappa' = [\kappa'_1, \kappa'_2, \dots, \kappa'_N]^T \quad (3)$$

[0262] For example, the target path $X_{t_1:t_s}$ of the bucket **6** from the current time t_1 to the modifiable start time t_s and the target path $X_{t_1:t_b}$ of the bucket **6** from the current time t_1 to the time t_b are expressed by the following equations (4) and (5).

[Math 3]

$$X_{t_1:t_s} = \{X_{t_1}, X_{t_1+1}, \dots, X_{t_s}\} \quad (4)$$

$$X_{t_1:t_b} = \{X_{t_1}, X_{t_1+1}, \dots, X_{t_b}\} \quad (5)$$

[0263] That is, in this example, the target path $X_{t_1:t_s}$ of the bucket **6** is defined as a set of positions X_t of the bucket **6** at each time t which are expressed discretely. For example, the position X_t of the bucket **6** at time t is expressed by the following equation (6) as a set of the attitude $\theta_{1,t}$ of the boom **4**, the attitude $\theta_{2,t}$ of the arm **5**, the attitude $\theta_{3,t}$ of the bucket **6**, and the attitude $\theta_{4,t}$ of the upper swivel body **3**.

[Math 4]

$$X_t = [\theta_{1,t}, \theta_{2,t}, \theta_{3,t}, \theta_{4,t}]^T \quad (6)$$

[0264] The attitude $\theta_{1,t}$ of the boom **4** is information representing, for example, the position (rod position) of the boom cylinder **7**. The attitude $\theta_{1,t}$ of the boom **4** may be information representing the attitude angle of the boom **4**. The data of the attitude $\theta_{1,t}$ of the boom **4** is acquired based on the output of the sensor **S7**.

[0265] The attitude $\theta_{2,t}$ of the arm **5** is, for example, information representing the position (rod position) of the arm cylinder **8**. The attitude $\theta_{2,t}$ of the arm **5** may be

information representing the attitude angle of the arm 5. The data of the attitude $\theta_{2,t}$ of the arm 5 is acquired based on the output of the sensor S8.

[0266] The attitude $\theta_{3,t}$ of the bucket 6 is, for example, information representing the position (rod position) of the bucket cylinder 9. The attitude $\theta_{3,t}$ of the bucket 6 may be information representing the attitude angle of the bucket 6. The data of the attitude $\theta_{3,t}$ of the bucket 6 is acquired based on the output of the sensor S9.

[0267] The attitude $\theta_{4,t}$ of the upper swivel body 3 is, for example, information representing the swivel angle of the upper swivel body 3. The data of the attitude $\theta_{4,t}$ of the upper swivel body 3 is acquired based on the output of the sensor S4 or the sensor S5.

[0268] In addition, the position X_t of the bucket 6 may include information on the velocity, acceleration, or jerk and the like of each of the boom 4, the arm 5, and the bucket 6.

[0269] For example, based on the shape h^t of the earth and the characteristic κ^t of the earth at the current time t_1 for each grid unit i of the observation target area TA, and the target path $X_{t1:ts}$ of the bucket 6 from the current time t_1 to the time t_s , the target material state prediction part 302B predicts the shape h^{ts} of the earth and the earth kts of the earth at the time t_s by using a function g corresponding to the trained model LM1. The shape h^{ts} of the earth and the characteristic kts of the earth at the time t_s is expressed by the following equation (7).

[Math 5]

$$(h^{ts}, \kappa^{ts}) = g(h^{t1}, \kappa^{t1}, X_{t1:ts}) \quad (7)$$

[0270] Similarly, based on the shape h^t of the earth and the characteristic κ^t of the earth at the current time t_1 for each grid unit i of the observation target area TA, and the target path $X_{t1:tb}$ of the bucket 6 from the current time t_1 to the time t_b , the target material state prediction part 302B predicts the shape h^{tb} of the earth and the characteristic κ^{tb} of the earth at the time t_b by using a function g . The shape h^{tb} of the earth and the characteristic κ^{tb} of the earth at the time t_b is expressed by the following equation (8).

[Math 6]

$$(h^{tb}, \kappa^{tb}) = g(h^{t1}, \kappa^{t1}, X_{t1:tb}) \quad (8)$$

[0271] The function g is based on, for example, a DNN.

[0272] When the processing in the step S204 is completed, the controller 30 proceeds to a step S206.

[0273] In the step S206, the operation planning part 302C determines the next predetermined operation (type) v^k that can be modified from among a plurality of predetermined operations corresponding to the current operation based on the current predetermined operation (type) v^{k-1} and the state of the earth predicted in the step S204.

[0274] For example, the operation planning part 302C determines the next predetermined operation (type) v^k that can be modified from among a plurality of predetermined operations corresponding to the current operation using the function f corresponding to the trained model LM2. The

predetermined operation v^k is expressed by the following equation (9) using the function f .

[Math 7]

$$v^k = f(v^{k-1}, h^{tb}, \kappa^{tb}) \quad (9)$$

[0275] The predetermined operations v^k and v^{k-1} are expressed by, for example, a one-hot vector.

[0276] The function f may be defined in the rule base instead of being given by the trained model LM2 as described above.

[0277] When the processing of the step S206 is completed, the controller 30 proceeds to a step S208.

[0278] In the step S208, the target path generation part 302D generates the target path of the bucket 6 in the predetermined operation of the shovel 100 after the time t_s , based on the prediction result (the state of the earth at the time t_s) of the process in step S204. Specifically, the target path generation part 302D generates the target path of the bucket 6 in the predetermined operation of the shovel 100 from the time t_s to the time t_b , and the target path of the bucket 6 in the predetermined operation of the shovel 100 after the time t_b until the predetermined timing.

[0279] When the process in the step S208 is completed, the controller 30 proceeds to a step S210.

[0280] In the step S210, the target path generation part 302D writes the data of the target path of the bucket 6 generated in the step S208 to a predetermined storage area (address) of the memory device 30B.

[0281] Thus, the operation control part 302E can access the latest data of the target path of the shovel 100 by accessing the predetermined address of the memory device 30B.

[0282] When the process in the step S210 is completed, the controller 30 ends the process of the present flowchart.

[0283] Thus, in this example, the controller 30 predicts a future shape of the earth, and determines the predetermined future operation of the shovel 100 based on the future shape of the earth. Thus, the controller 30 can generate the target path of the bucket 6 based on the predicted future shape of the earth and the determined predetermined future operation.

<Process Related to Target Path Generation (Sub-Flow)>

[0284] FIG. 15 is a sub-flowchart schematically illustrating an example of processing related to path generation of the bucket 6. FIG. 16 is a drawing illustrating an example of cost conditions and operation parameters corresponding to a plurality of sections of a shovel 100 excavation operation.

[0285] The sub-flowchart of FIG. 15 corresponds to the process of the step S208 in FIG. 13.

[0286] As shown in FIG. 15, in a step S302, the target path generation part 302D selects a constraint function related to the path of the bucket 6 corresponding to (type of) a predetermined operation of the shovel 100. At this time, the predetermined operation of the shovel 100 is a predetermined operation of the shovel 100 performed at time t_s and a predetermined operation of the shovel 100 starting from time t_b . Specifically, for each predetermined operation, the target path generation part 302D selects a constraint function related to the path of the bucket 6 corresponding to the predetermined operation.

[0287] The constraint function (constraint condition) includes, for example, constraints related to the moving range, speed, and acceleration of the boom cylinder 7, the arm cylinder 8, and the bucket cylinder 9. Further, the constraint function (constraint condition) may include a constraint condition for avoiding collision between an obstacle around the shovel 100 and the bucket 6. The obstacle around the shovel 100 includes, for example, a person, a working vehicle, another working machine, a feature (e.g., fence, utility pole), and the like, and can be recognized based on the output of the sensor 40.

[0288] When the processing of the step S302 is completed, the controller 30 proceeds to a step S304.

[0289] In the step S304, the target path generation part 302D selects an objective function (cost function) related to the path of the bucket 6 corresponding to (type of) the predetermined operation of the shovel 100. At this time, the predetermined operation of the shovel 100 is the predetermined operation of the shovel 100 performed at time t_s and the predetermined operation of the shovel 100 starting from time t_b , as in the case of step S302. Specifically, for each predetermined operation, the target path generation part 302D selects a cost function related to the path of the bucket 6 corresponding to the predetermined operation.

[0290] For example, as shown in FIG. 16, the excavation operation as the predetermined operation of the shovel 100 is divided into the operation sections of approach, penetration, horizontal excavation, and scooping up.

[0291] “Approach” is an operation section in which the bucket 6 is brought close to the ground in order to penetrate the ground. “Penetration” is an operation section in which the cutting edge of the bucket 6 is brought into contact with the ground after the operation section of approach and the bucket 6 is penetrated to a certain depth of the ground. “Horizontal excavation” is an operation section in which the bucket 6 is moved approximately horizontally after the operation section of penetration. “Scooping up” is an operation section in which earth are stored in the bucket 6 after horizontal excavation and the earth are scooped up on the ground.

[0292] In this example, cost functions related to the speed and acceleration of the bucket 6, the travel time of the bucket 6, and the like are defined over all the operation sections from “approach” to “scooping up”.

[0293] In addition, cost functions related to the position of the cutting edge of the bucket 6, the angle of the cutting edge to a predetermined reference (e.g., a horizontal plane), and the path of the cutting edge are defined at the end of one operation section among “approach”, “penetration”, “horizontal excavation”, and “scooping up”, and in one operation section among the above.

[0294] When the processing of the step S304 is completed, the controller 30 proceeds to a step S306.

[0295] In the step S306, the target path generation part 302D estimates an operation parameter that defines the target path of the bucket 6 in a predetermined operation of the shovel 100 using the trained model LM3 based on the prediction result (state of the earth of the target material at time t_s , t_b) in the step S302.

[0296] For example, as shown in FIG. 16, operation parameters q_1 to q_4 that define the position of the cutting edge of the bucket 6, and operation parameters β_{12} , β_{23} , and ρ_4 that define the angle of the cutting edge of the bucket 6 to a predetermined reference are defined.

[0297] The operation parameter q_1 is an operation parameter representing the position of the cutting edge of the bucket 6 at the end of “approach” and at the beginning of “penetration”. In this example, at the end of “approach” and at the beginning of “penetration”, a cost function corresponding to the condition for determining that the position of the cutting edge of the bucket 6 matches the position of the operation parameter q_1 is defined.

[0298] The operation parameter q_2 is an operation parameter representing the position of the cutting edge of the bucket 6 at the end of “penetration” and at the beginning of “horizontal excavation”. In this example, at the end of “penetration” and at the beginning of “horizontal excavation”, a cost function corresponding to the condition for determining that the position of the cutting edge of the bucket 6 matches the position of the operation parameter q_2 is defined.

[0299] In this example, a cost function corresponding to the condition for determining that the position (path) of the cutting edge is on the straight line defined by the operation parameters q_1 and q_2 in the operation section of “penetration” is defined.

[0300] The operation parameter q_3 is an operation parameter representing the position of the cutting edge of the bucket 6 at the end of “horizontal excavation” and at the beginning of “scooping up”. In this example, the cost function corresponding to the condition for determining that the position of the edge of the bucket 6 and the position of the operation parameter q_3 match at the end of “horizontal excavation” and at the beginning of “scooping up” is defined.

[0301] In this example, the cost function corresponding to the condition for determining that the position (path) of the edge is on the straight line defined by the operation parameters q_2 and q_3 in the operation section of “horizontal excavation” is defined.

[0302] The operation parameter q_4 is an operation parameter representing the position of the edge of the bucket 6 at the end of “scooping up”. In this example, the cost function corresponding to the condition for determining that the position of the edge of the bucket 6 and the position of the operation parameter q_4 match at the end of “scooping up” is defined.

[0303] The operation parameter ρ_{12} is an operation parameter representing the angle of the edge of the bucket 6 with respect to the predetermined reference in the operation section of “penetration”. In this example, the constraint function and cost function corresponding to the condition for determining that the angle of the edge of the bucket 6 with respect to the predetermined reference and the angle of the operation parameter ρ_{12} match in the operation section of “penetration” are defined.

[0304] The operation parameter ρ_{23} is an operation parameter representing the angle of the cutting edge of the bucket 6 with respect to a predetermined reference in the operation section of “horizontal excavation”. In this example, in the operation section of “horizontal excavation”, the constraint function and the cost function, corresponding to the condition for determining that the angle of the cutting edge of the bucket 6 with respect to a predetermined reference matches the angle of the operation parameter ρ_{23} , are defined.

[0305] The operation parameter ρ_4 is an operation parameter representing the angle of the cutting edge of the bucket 6 with respect to a predetermined reference at the end of

“scooping up”. In this example, the constraint function and the cost function corresponding to the condition for determining that the angle of the cutting edge of the bucket 6 with respect to a predetermined reference matches the angle of the operation parameter p_4 at the end of “scooping up” are defined.

[0306] When the processing in the step S306 is completed, the controller 30 proceeds to a step S308.

[0307] In the step S308, the target path generation part 302D calculates the target path of the bucket 6 in the predetermined operation of the shovel 100 based on the constraint function and the objective function selected in the steps S302 and S304, and the operation parameter of the estimation result in the step S306. Specifically, the target path generation part 302D calculates the target path of the bucket 6 in the predetermined operation of the shovel 100 by solving a constrained nonlinear optimization problem defined by the constraint function and the objective function using a predetermined solver.

[0308] When the processing of the step S308 is completed, the processing of the present sub-flowchart is terminated.

[0309] Thus, in this example, the controller 30 can generate the target path of the shovel 100 by using the constraint function and the objective function corresponding to the predetermined operation of the shovel 100.

[0310] In this example, the path of the bucket 6 is represented by a relatively small number of operation parameters. Then, the controller 30 can infer the operation parameters which define the path of the bucket 6 by using the trained model LM3 based on the prediction result of the state of the earth of the future target material.

<Processing Related to Operation Control of Shovel>

[0311] FIG. 17 is a flowchart schematically illustrating an example of processing related to operation control of the shovel 100.

[0312] This flowchart is repeatedly performed for each predetermined processing cycle during performance of the autonomous operation of the shovel 100, for example.

[0313] As shown in FIG. 17, in a step S402, the operation control part 302E reads the latest data representing the target path of the bucket 6 in the predetermined operation of the shovel 100 from the predetermined address of the memory device 30B. This data is the data registered in the step S210 of FIG. 13.

[0314] When the processing in the step S402 is completed, the controller 30 proceeds to a step S404.

[0315] In the step S404, the operation control part 302E controls the operation of the shovel 100 based on the data of the target path of the bucket 6 in the predetermined operation of the shovel 100 read in the step S402. Specifically, the operation control part 302E controls the operation of the shovel 100 while outputting a control command to the hydraulic controlling valve 31 so that the bucket 6 moves along the target path corresponding to the data read in the step S402.

[0316] When the process of the step S404 is completed, the process of the present flowchart ends.

[0317] Thus, in this example, the controller 30 can control the operation of the shovel 100 so as to move along the target path of the predetermined operation of the shovel 100.

[Specific Example of Trained Model Generation Method]

[0318] Next, a specific example of a method for generating a trained model LM1 will be described.

[0319] In this example, a method for generating a function g corresponding to the trained model LM1 used in the process of FIG. 13 will be described.

[0320] Training data c^j ($j=1$ to L (an integer of 2 or more)) of a training data set D generated by the training data generation part 2004A is expressed, for example, by the following equation (10).

[Math 8]

$$c^j = \{h_r^j, \kappa_r^j, \hat{h}^j, \hat{\kappa}^j, X^j\} \quad (10)$$

[0321] The training data c^j is a combination of input data h^j , κ^j , and X^j , and the shape h_r^j of the earth and the characteristic κ_r^j of the earth at the time of completion of the path X^j as the ground truth. The input data h^j , κ^j , and X^j correspond to the input data of the function g in equations (7) and (8).

[0322] As described above, the training data set D may be generated from the log acquired by the log acquisition part 2001, the log acquired by the simulator part 2002, or both logs.

[0323] In the simulator part 2002, for example, as described above, a particle simulation such as DEM is adopted, and the height h_r^j of the earth is acquired by ray tracing of the shape sensor such as LIDAR which is virtually arranged with respect to the position of the particle.

[0324] As described above, the training data set D may include a base training data set generated from logs acquired by the simulator part 2002 and a fine-tuning training data set generated from logs acquired by the log acquisition part 2001. In this case, the number of training data included in the fine-tuning training data set may be relatively small.

[0325] As shown in the following equation (11), the function g has a parameter W , and the machine learning part 2005A performs machine learning in such a manner that the parameter W is optimized by the training data set D .

[Math 9]

$$(h^j, \kappa^j) = g(\hat{h}^j, \hat{\kappa}^j, X^j; w) \quad (11)$$

[0326] For example, the function g corresponding to the trained model LM1 is generated by optimizing the parameter W so that the loss function $E(W)$ in the following equation (12) is minimized.

[Math 10]

$$E(w) = - \sum_j \log N(h_r^j | h^j, \kappa^j; W) \quad (12)$$

[0327] Thus, the information processing device 200 generates the training data set D including the training data c^j , and can generate the function g corresponding to the trained model LM1 by machine learning based on the training data set D .

[Action]

[0328] Next, the action of the working machine, the information processing device, and the program according to the present embodiment will be described.

[0329] In the present embodiment, the working machine includes an operation planning part. The working machine is, for example, the above-described shovel 100. The operation planning part is, for example, the above-described operation planning part 302C. Specifically, the operation planning part determines the future operation of the working machine from among a plurality of according operations to the performance status of the operation of the working machine.

[0330] In the present embodiment, the information processing device may include an operation planning part. The information processing device may be, for example, the controller 30, the information processing device 200, or the remote control support device 400 described above.

[0331] In the present embodiment, the program may cause the information processing device to perform an operation planning step in order to achieve the function of the operation planning part. Specifically, in the operation planning step, the future operation of the working machine is determined from among a plurality of operations according to the performance status of the operation of the working machine. The operation planning step is, for example, the step S206 described above.

[0332] Thus, the working machine or the like can determine the future operation of the working machine from among a plurality of operations in consideration of the state of the target material which changes according to the performance status of the operation of the working machine. Therefore, the working machine can achieve a more appropriate operation.

[0333] In the present embodiment, the working machine and the information processing device may include a prediction part. The prediction part is, for example, the target material state prediction part 302B described above. Specifically, the prediction part may predict a future state of the target material according to the performance status of the operation of the working machine. The operation planning part may determine the future operation of the working machine from among the plurality of operations, based on the prediction result of the future state of the target material by the prediction part.

[0334] In the present embodiment, the program may cause the information processing device to perform the prediction step in order to achieve the function of the prediction part. Specifically, the prediction step may predict the future state of the target material according to the performance state of the operation of the working machine. The operation planning step may decide the future operation of the working machine from among a plurality of operations based on the prediction result of the future state of the target material by the prediction part.

[0335] Thus, the working machine or the like can determine the future operation of the working machine from among a plurality of operations in accordance with the future state of the target material predicted according to the performance status of the operation of the working machine.

[0336] In the present embodiment, the prediction part may predict the future state of the target material, based on the path of the work part by the operation of the working machine.

[0337] Thus, the working machine or the like can predict the future state of the target material considering the change of the state of the target material which changes in accordance with the path of the work part.

[0338] In the present embodiment, the prediction part may predict a state of the target material after completion of a current operation of the working machine or a next operation after the current operation, based on the path of the work part by the current operation of the working machine or a next scheduled operation. Then, the operation planning part determines a next operation after the current operation of the working machine or a further next operation after the next scheduled operation from among the plurality of operations, based on the prediction result of the future state of the target material by the prediction part.

[0339] Thus, the working machine or the like can determine the next operation according to the prediction result of the work state after the completion of the operation under performance of the working machine. Furthermore, the working machine or the like can determine the next operation after the next operation scheduled to be performed according to the prediction result of the work state after the completion of the next operation.

[0340] In addition, in the present embodiment, the working machine or the like may include a generation part and a control part. The generation part is, for example, the target path generation part 302D described above. The control part is, for example, the operation control part 302E described above. Specifically, the generation part may generate the path of the work part by the operation of the working machine, based on the state of the target material. The control part may control the operation of the working machine so that the work part moves along the path generated by the generation part. The prediction part may predict the future state of the target material, based on the current state of the target material and the path of the work part generated by the generation part.

[0341] In the present embodiment, the program may cause the information processing device to perform the generation step and the control step in order to achieve the functions of the generation part and the control part. The generation step is, for example, the step S208 described above. The control step is, for example, the step S404 described above. Specifically, in the generation step, the path of the work part by the operation of the working machine may be generated based on the state of the target material. In the control step, the operation of the working machine may be controlled so that the work part moves along the path generated in the generation step. In the prediction step, the future state of the target material may be predicted based on the current state of the target material and the path of the work part generated in the generation step.

[0342] Thus, the working machine or the like can predict the future state of the target material based on the path of the work part generated for automatic operation of the working machine.

[0343] In the present embodiment, the prediction part may predict the state of the target material after the elapse of a predetermined time. The generation part may generate the path of the work part after the elapse of the predetermined time, based on the prediction result of the state of the target material after the elapse of the predetermined time by the prediction part.

[0344] Thus, the working machine or the like can generate the path of the work part after the elapse of the predetermined time considering, for example, the delay time from the instruction to control the operation of the working machine to the actual control of the operation.

[0345] Further, in the present embodiment, the generation part may generate the path of the work part, based on measurement data of the state of the target material, by using an objective function and a constraint function defined for each of the plurality of operations.

[0346] Thus, the working machine or the like can generate the path of the work part.

[0347] Further, in the present embodiment, the path of the work part may be expressed by a predetermined number, which is two or greater, of parameters for each of the plurality of operations. Then, the generation part may generate the path of the work part by determining a predetermined number of parameters, based on the state of the target material.

[0348] Thus, the working machine or the like can generate the path of the work part.

[0349] In the present embodiment, the state of the target material may include at least one of the shape and the characteristic of the earth on the surface of the target material.

[0350] Thus, the working machine or the like can predict the future state of the target material in accordance with the shape and the characteristic of the earth of the target material.

[0351] In the present embodiment, the working machine or the like may include a notification part for notifying the operator of the operation determined by the operation planning part.

[0352] In the present embodiment, the program may cause the support device to perform the operation planning step and the notification step. The support device is, for example, the support device 150 and the remote control support device 400. Specifically, in the operation planning step, the future operation of the working machine may be decided from among a plurality of operations according to the performance status of the operation of the working machine. In the notification step, the operation decided in the operation planning step may be notified to the operator of the working machine.

[0353] Thus, the working machine or the like can notify the operator of a more appropriate type of operation of the working machine to the operator who operates the working machine. Therefore, the operator can operate the working machine more appropriately. Moreover, the working machine or the like can notify the operator in advance of a more appropriate type of operation of the working machine to the operator who operates the working machine. Therefore, the operator can start the next operation immediately after the completion of a certain operation, and as a result, the work efficiency of the working machine can be improved.

[0354] Further, the present disclosure is not limited to these embodiments, but various variations and modifications may be made without departing from the scope of the present disclosure.

What is claimed is:

1. An information processing device comprising an operation planning part configured to determine a future operation

of a working machine from among a plurality of operations according to a performance status of an operation of the working machine.

2. A working machine comprising the information processing device according to claim 1, and an operation planning part configured to determine a future operation of the working machine from among a plurality of operations according to a performance status of an operation of the working machine.

3. The working machine according to claim 2, further comprising a prediction part configured to predict a future state of a target material according to the performance status of the operation of the working machine,

the operation part wherein planning is configured to determine the future operation of the working machine from among the plurality of operations, based on a prediction result of the future state of the target material by the prediction part.

4. The working machine according to claim 3, wherein the prediction part is configured to predict the future state of the target material, based on a path of a work part by an operation of the working machine.

5. The working machine according to claim 4, wherein: the prediction part is configured to predict a state of the target material after completion of a current operation of the working machine or a next operation after the current operation, based on the path of the work part by the current operation of the working machine or a next scheduled operation; and

the operation planning part is configured to determine the next operation after the current operation of the working machine or a further next operation after the next scheduled operation from among the plurality of operations, based on the prediction result of the future state of the target material by the prediction part.

6. The working machine according to claim 4, further comprising:

a generation part configured to generate a path of the work part by an operation of the working machine, based on a state of the target material; and

a control part configured to control an operation of the working machine so that the work part moves along the path generated by the generation part,

wherein the prediction part is configured to predict the future state of the target material, based on a current state of the target material and the path of the work part generated by the generation part.

7. The working machine according to claim 6, wherein: the prediction part is configured to predict the state of the target material after a lapse of a predetermined time; and

the generation part is configured to generate the path of the work part after the lapse of the predetermined time, based on the prediction result of the state of the target material after the lapse of the predetermined time by the prediction part.

8. The working machine according to claim 6, wherein the generation part is configured to generate the path of the work part, based on measurement data of the state of the target material, by using an objective function and a constraint function defined for each of the plurality of operations.

9. The working machine according to claim 6, wherein: the path of the work part is expressed by a predetermined number of parameters for each of the plurality of operations, the predetermined number being two or greater; and the generation part is configured to generate the path of the work part by determining the predetermined number of parameters, based on the state of the target material.
10. The working machine according to claim 3, wherein the state of the target material includes at least one of a shape and a characteristic of earth on a surface of the target material.
11. The working machine according to claim 2, further comprising a notification part configured to notify an operator of the future operation determined by the operation planning part.
12. A non-transitory computer-readable recording medium having a program embodied therein for causing an information processing device to determine a future operation of a working machine from among a plurality of operations according to a performance status of an operation of the working machine.
13. A non-transitory computer-readable recording medium having a program embodied therein for causing a support device to perform:
- determining a future operation of a working machine from among a plurality of operations according to a performance status of an operation of the working machine; and
 - notifying an operator of the determined future operation.
14. The working machine according to claim 2, wherein the plurality of operations are of different types, and each type is predetermined.
15. The working machine according to claim 3, further comprising a storage part configured to store a trained model for outputting one operation from among the plurality of operations, when the state of the target material is an input, wherein the operation planning part is configured to determine the future operation of the working machine

- from among the plurality of operations by using the trained model, based on the prediction result of the future state of the target material by the prediction part.
16. The working machine according to claim 3, wherein: a candidate operation capable of transitioning as a next operation within the plurality of operations, starting from a target operation by each of the plurality of operations, and a transition condition for transitioning to the candidate operation are predetermined; and the operation planning part is configured to determine the future operation of the working machine from among the plurality of operations by using the transition condition, based on the prediction result of the future state of the target material by the prediction part.
17. The working machine according to claim 3, wherein: the prediction part is configured to predict a state of the target material at a predetermined time in future, based on the operation status of the working machine; and the operation planning part is configured to determine a future operation starting from the predetermined time, based on the prediction result of the state of the target material at the predetermined time in the future by the prediction part.
18. The working machine according to claim 2, wherein the operation planning part is configured to determine an undetermined future operation of the working machine from among a plurality of operations according to the performance status of the operations of the working machine.
19. The working machine according to claim 3, further comprising a storage part configured to store a trained model for outputting one operation from among the plurality of operations, when information concerning the performance status of the operation of the working machine is an input, wherein the prediction part is configured to predict the future state of the target material, based on the information concerning the performance status of the operation of the working machine, using the trained model.

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